77

Global Considerations in Kidney Disease: Africa

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CHAPTER OUTLINE

KEY POINTS

- The growing burden of renal disease in Africa is driven by the high rates of infectious diseases, noncommunicable diseases, pregnancy-related diseases, and injuries.
- In most African countries, financial and human resources are inadequate to meet this challenge.
- Most patients with end-stage renal disease cannot access renal replacement therapy and the lack of access to acute dialysis results in many potentially preventable deaths due to acute kidney injury.
- Positive initiatives that provide hope for the future include the training of nephrology healthcare professionals via the ISN and ISPD Fellowship programs, increased awareness of kidney disease resulting from World Kidney Day activities, and the Saving Young Lives and 0by25 projects, which are improving the outcomes for patients with acute kidney injury.

INTRODUCTION

Africa is the world's second-largest and second-most-populous continent, including 55 countries and diverse ethnic groups, religions and languages. Its population in mid-2017 was estimated at 1.3 billion and it is projected to reach 2.6 billion by 2050, at which time it will account for most of the global population increase (Fig. 77.1).[1,2] Half of the world's population growth is expected to be concentrated in nine countries, and this list includes five countries from Africa: Nigeria, the Democratic Republic of the Congo, Ethiopia, the United Republic of Tanzania, and Uganda.[2]

The African population is very young, with 41% of people being under the age of 15 years old and only 3% being 65 years or older, compared with the world averages of 26% and 9%, respectively.[1] In Western Africa, the life expectancy of men is 55 years and 57 years for women, both of which are the lowest of any world region.[1] These findings are in stark contrast to many other regions of the world, where lower fertility rates and increasing life expectancy have contributed to the aging of populations. Africa's youth population is predicted to rise from 20% in 2017 to 35% of the world youth population in 2050 (Fig. 77.2).

Globally, the highest birth rates are seen in African countries, with the highest average lifetime births per woman being recorded in Niger (7.3), Chad (6.4), and Somalia (6.4).[1] Child marriages and adolescent childbearing are common in many countries of sub-Saharan Africa,[3] and pose serious consequences for the health of young girls and their offspring. Early childbearing also limits their educational and employment opportunities.

Table 77.1 summarizes the population estimates and the economic, and development indicators of African countries. Africa is by far the world's poorest region and is currently recovering from a sharp economic slowdown experienced over the last two years. There are improved outputs from the mining and agriculture sectors, which are boosting

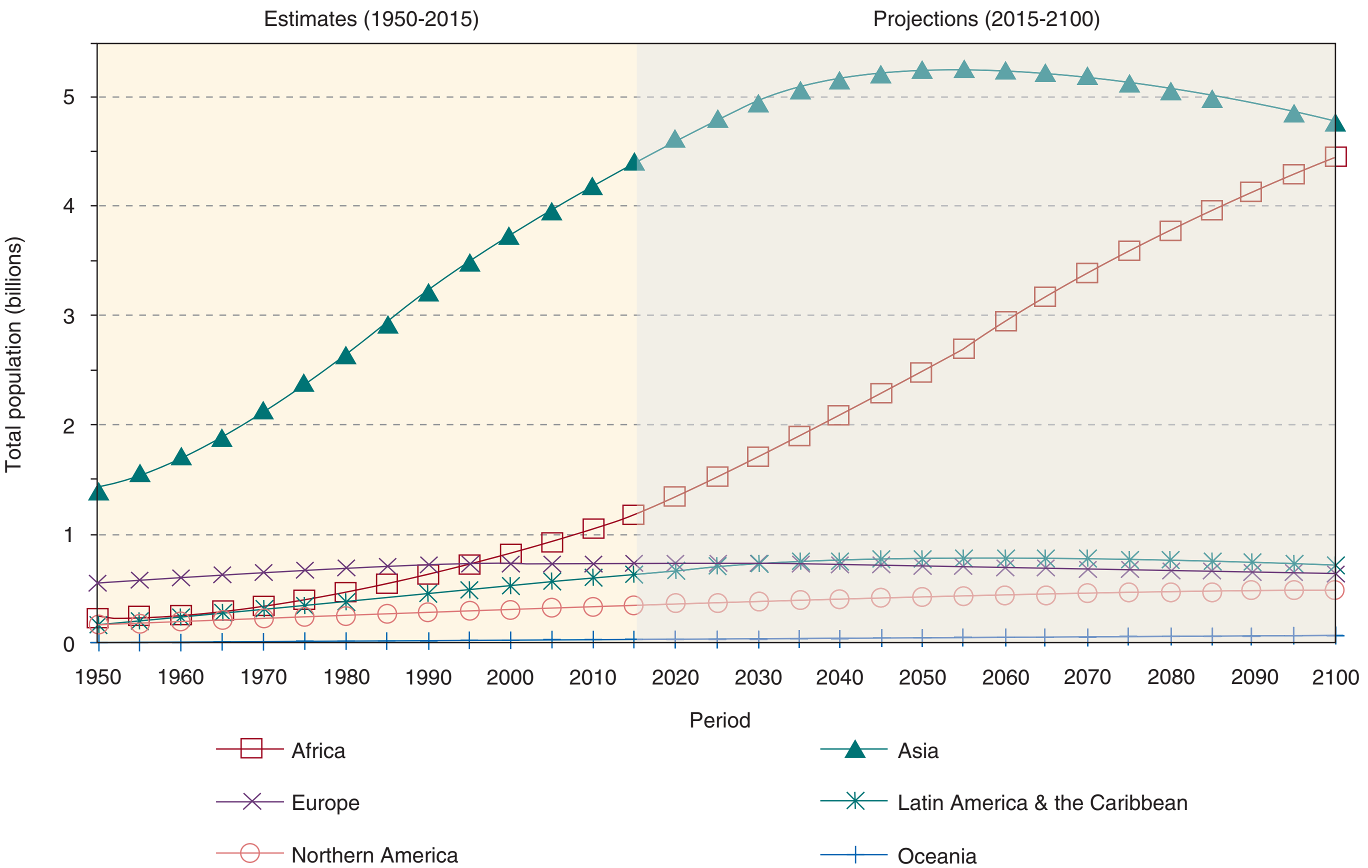

Fig. 77.1 World population by region, with projections to 2100. (From World Population Prospects: *The 2017 revision, key findings and advance tables: United Nations, Department of Economic and Social Affairs, Population Division;* 2017.)

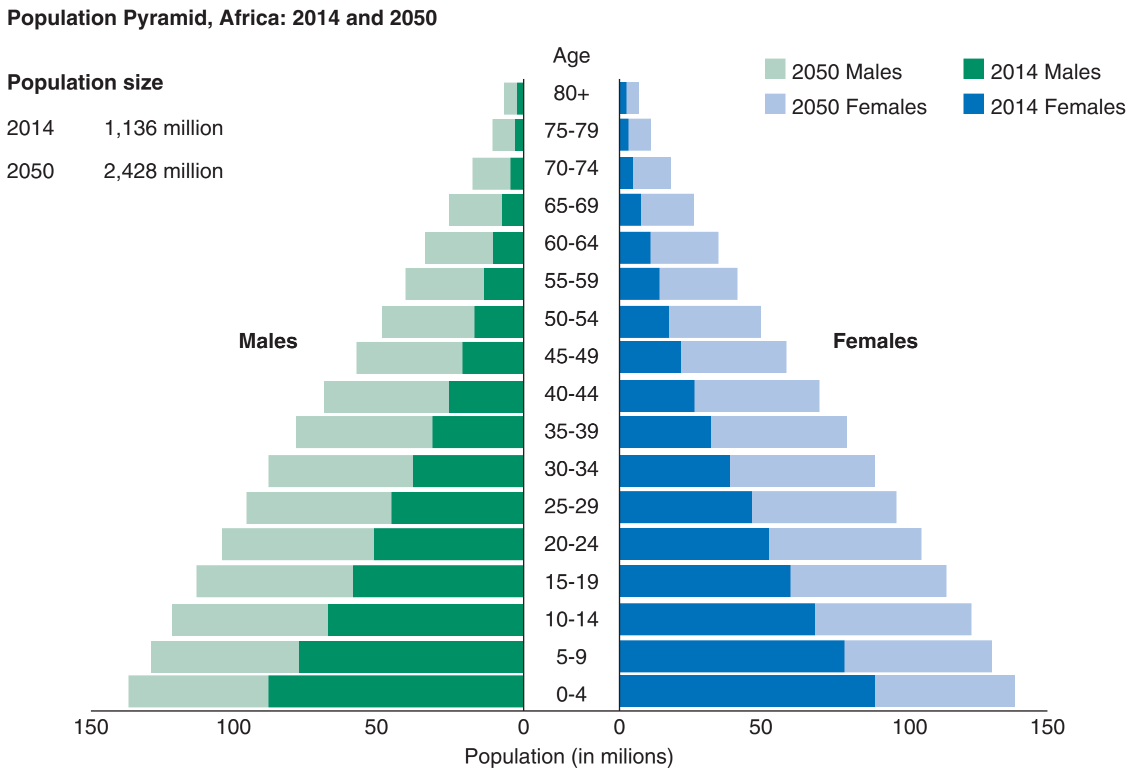

Fig. 77.2 Population pyramid for Africa; 2014 and 2050. (From *Noncommunicable diseases among young people in Africa*. Washington, DC: Population Reference Bureau; 2015.)

Table 77.1 Population Estimates, Economic and Development Indicators of African Countries

Subregion and Country	Total Population (Thousands)			GNIPC	Life Exp	MMR	Under 5	New HIV
	2017	2030	2050					
AFRICA	1 256 268	1 703 538	2 527 557					
Eastern Africa	422 036	587 330	888 129					
Burundi	10 864	15 799	25 762	280	52.2	712	81.7	0.1
Comoros	814	1 062	1 463	760	55.9	335	73.5	
Djibouti	957	1 133	1 308	[a] 1030	55.8	229	65.3	1.1
Eritrea	5 069	6 718	9 607	[a] 520	55.9	501	46.5	0.2
Ethiopia	104 957	139 620	190 870	660	56.1	353	59.2	
Kenya	49 700	66 960	95 467	1 380	55.6	510	49.4	2.3
Madagascar	25 571	35 592	53 803	400	56.9	353	49.6	0.2
Malawi	18 622	26 578	41 705	320	51.2	634	64	4.5
Mauritius	1 265	1 287	1 221	9 760	66.8	53	13.5	0.4
Mayotte	253	344	495					
Mozambique	29 669	42 439	67 775	480	49.6	489	78.5	7.4
Reunion	877	957	1 014					
Rwanda	12 208	16 024	21 886	700	56.6	290	41.7	1.1
Seychelles	95	98	97	15 410	65.5		13.6	
Somalia	14 743	21 535	35 852		47.8	732	136.4	0.5
South Sudan	12 576	17 254	25 366	820	49.9	789	92.6	2.6
Uganda	42 863	63 842	105 698	660	54.0	343	54.6	6
United Republic of Tanzania	57 310	83 702	138 082	900	54.1	398	48.7	2.6
Zambia	17 094	24 859	41 001	1300	53.6	224	64	7.5
Zimbabwe	16 530	21 527	29 659	940	52.3	443	70.7	9.2
Middle Africa	163 495	237 771	384 005					
Angola	29 784	44 712	76 046	3 440	45.8	477	156.9	2.1
Cameroon	24 054	32 980	49 817	1 200	50.3	596	87.9	3.8
Central African Republic	4 659	6 124	8 851	370	45.9	882	130.1	2.7
Chad	14 900	21 460	33 636	720	46.1	856	138.7	1.5
Congo	5 261	7 319	11 510	1 710	56.6	442	45	1.4
Democratic Republic of the Congo	81 340	120 443	197 404	420	51.7	693	98.3	0.6
Equatorial Guinea	1 268	1 871	2 845	6 550	51.2	342	94.1	2.9
Gabon	2 025	2 594	3 516	7 210	57.2	291	50.8	1.4
Sao Tome and Principe	204	268	380	1730	59.1	156	47.3	0.1
Northern Africa	233 604	285 204	359 905					
Algeria	41 318	48 822	57 437	4 270	66.0	140	25.5	<0.1
Egypt	97 553	119 746	153 433	3 460	62.2	33	24	<0.1
Libya	6 375	7 342	8 124	[a] 4730	63.8	9	13.4	
Morocco	35 740	40 874	45 660	2850	64.9	121	27.6	0.1
Sudan	40 533	54 842	80 386	2140	55.9	311	70.1	0.2
Tunisia	11 532	12 842	13 884	3 690	66.7	62	14	<0.1
Southern Africa	65 143	74 786	85 800					
Botswana	2 292	2 800	3 421	6 610	56.9	129	43.6	14
Lesotho	2 233	2 608	3 203	1 210	46.6	487	90.2	20.1
Namibia	2 534	3 246	4 339	4 680	57.5	265	45.4	9.1
South Africa	56 717	64 466	72 755	5 480	54.5	138	40.5	12.7
Swaziland	1 367	1 666	2 081	2830	50.9	389	60.7	18.9
Western Africa	371 990	518 446	809 719					
Benin	11 176	15 628	23 930	820	52.5	405	99.5	0.6
Burkina Faso	19 193	27 382	43 207	640	52.6	371	88.6	0.5
Cabo Verde	546	635	734	2970	64.4	42	24.5	0.9
Cote d’Ivoire	24 295	33 337	51 375	1520	47.0	645	92.6	2.1
Gambia	2 101	3 001	4 562	440	53.8	706	68.9	1.1
Ghana	28 834	37 294	51 270	1 380	55.3	319	61.6	0.7
Guinea	12 717	17 631	26 852	490	51.7	679	93.7	1.1
Guinea-Bissau	1 861	2 493	3 603	620	51.5	549	92.5	2.5
Liberia	4 732	6 495	9 804	370	52.7	725	69.9	0.6
Mali	18 542	27 057	44 020	750	51.1	587	114.7	1.3
Mauritania	4 420	6 077	8 965	1 120	55.1	602	84.7	0.4
Niger	21 477	34 994	68 454	370	54.2	553	95.5	<0.1
Nigeria	190 886	264 068	410 638	2450	47.7	814	108.8	2

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Table 77.1 Population Estimates, Economic and Development Indicators of African Countries (Cont'd)

	Total Population (Thousands)							
Subregion and Country	2017	2030	2050	GNIPC	Life Exp	MMR	Under 5	New HIV
Saint Helena	4	4	4					
Senegal	15 851	22 123	34 031	950	58.3	315	47.2	<0.1
Sierra Leone	7 557	9 720	12 972	490	44.4	1360	120.4	0.7
Togo	7 798	10 507	15 298	540	52.8	368	78.4	1

[a]GNIPC data for Djibouti from 2005, Eritrea 2011, and Libya 2011.

GNIPC, Gross national income per capita for 2016 by the Atlas method; *Life Exp*, life expectancy at birth; *MMR*, maternal mortality rate per 100 000 live births; *Under 5*, mortality rate per 1000 children under the age of 5 years; *HIV infections*, new human immunodeficiency virus infections per 1000 adults aged 15–49 years.

Adapted from *Noncommunicable diseases among young people in Africa.* Washington, DC: Population Reference Bureau; 2015; the Global Health Observatory; and the World Bank World Development Indicators database.

Table 77.2 Examples of Noncommunicable Diseases Which Are Linked to Poverty

	Condition	Risk Factors Related to Poverty
Cardiovascular	Hypertension	Treatment gap
	Pericardial disease	Tuberculosis
	Rheumatic valvular disease	Streptococcal diseases
	Cardiomyopathies	Human immunodeficiency virus (HIV), other viruses, pregnancy
	Congenital heart disease	Maternal rubella, micronutrient deficiency, treatment gap
Respiratory	Chronic pulmonary disease	Indoor air pollution, tuberculosis, schistosomiasis, treatment gap
Endocrine	Diabetes mellitus	Malnutrition
	Hyperthyroidism and hypothyroidism	Iodine deficiency
Neurological	Epilepsy	Meningitis, malaria
	Stroke	Rheumatic mitral stenosis, endocarditis, malaria, HIV
Renal	Chronic kidney disease	Streptococcal disease
Musculoskeletal	Chronic osteomyelitis	Bacterial infection, tuberculosis
	Musculoskeletal injury	Trauma

Modified from Marquez PV, Farrington JL. *The challenge of non-communicable diseases and road traffic injuries in Sub-Saharan Africa: an overview.* Washington, DC: The World Bank; 2013.

economic activity but investment growth remains low and productivity growth is falling. The gross domestic product (GDP) growth in the region was expected to increase from 1.3% in 2016 to 2.4% in 2017, mainly led by the continent's largest economies: Nigeria, South Africa, and Angola.[4] However, this moderate improvement will only translate into slow gains in per capita income and will likely be insufficient to promote shared prosperity or accelerate poverty reduction. The proportion of total expenditure on health as a proportion of GDP is highly variable across the region, where universal health coverage is uncommon (Fig. 77.3). Poverty, overcrowding, and unhealthy environments persist as common social conditions that affect the occurrence of many diseases and the care available to address them (Table 77.2).

Africa labors under the quadruple burdens of communicable, noncommunicable, perinatal and maternal, and injury-related disorders. These factors are driving the current epidemic of kidney disease. Acute kidney injury (AKI) often develops in the setting of infectious diseases, complicated pregnancies, or injuries. Chronic kidney disease (CKD) may be related to human immunodeficiency virus (HIV) or other infections and is a frequent complication of noncommunicable diseases (NCDs) such as diabetes mellitus and hypertension.

Although communicable diseases have long been the leading causes of death and disease burden in Africa, the pattern of disease is changing as a result of rising incomes, population growth and aging, globalization, rapid urbanization, and Westernized lifestyles. NCDs are fast becoming the leading cause of death throughout Africa.[6] NCDs are emerging in both rural and urban areas, and most prominently in poor people living in urban settings, resulting in increasing pressure on acute and chronic healthcare services.[7] In North African countries, NCDs are already responsible for more than three-quarters of all deaths. In sub-Saharan Africa, NCDs account for more than 25 percent of deaths in 80 percent of the countries and by 2030, NCDs will be the leading cause of death even in sub-Saharan Africa. This is already the case in countries such as Mauritius and the Seychelles and in some populations, such as those over the age of 45 years.[8]

Four key risk behaviors for NCDs—tobacco and alcohol use, physical inactivity, and unhealthy diet—are on the rise among young Africans. These risk factors will need to be addressed to shift the projected trajectory of NCDs in Africa.[6]

Overweight and obesity, especially among girls, are emerging as critical public health issues even where undernutrition still remains a problem. Urbanization and globalization have

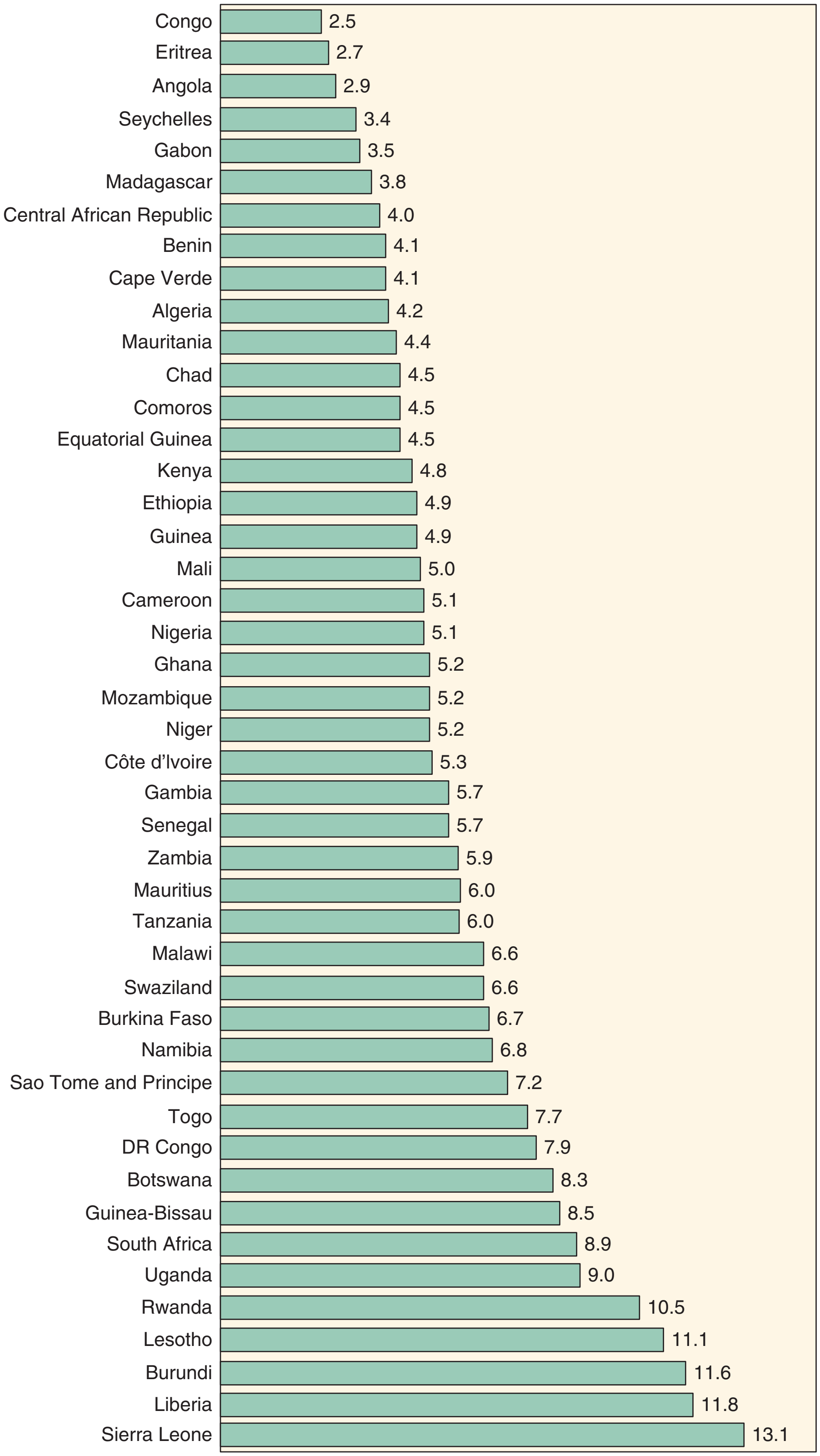

Fig. 77.3 Total expenditure on health as percentage of gross domestic product in the African region, 2010.

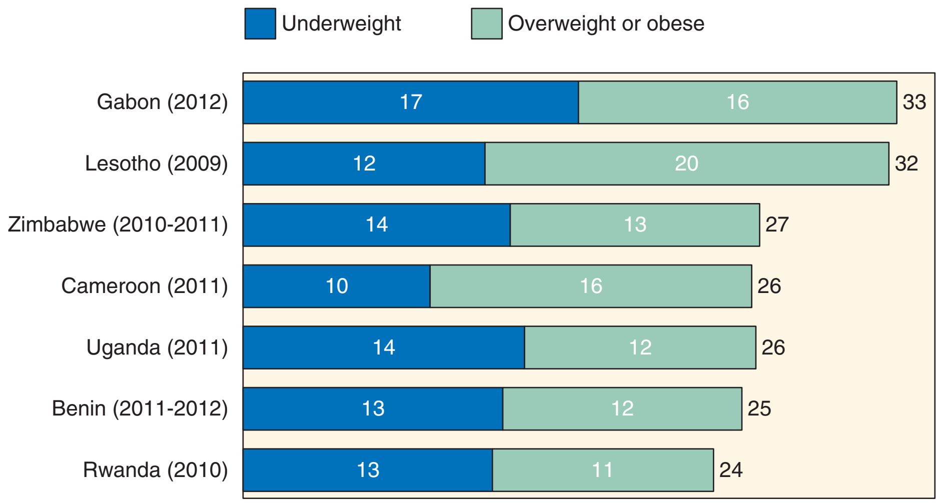

Fig. 77.4 Overweight/obesity or underweight in young women 15–19 years old. (From *Noncommunicable diseases among young people in Africa.* Washington, DC: Population Reference Bureau; 2015.)

led to sedentary lifestyles and diets filled with calorie-dense, processed foods that are high in saturated fat, sodium, and sugar. These shifts are leading to a rise in NCDs such as type 2 diabetes, cardiovascular disease, stroke, and certain cancers.[6] Fig. 77.4 illustrates the percentage of young women with over- or undernutrition in selected African countries.

Africans are using more tobacco and are starting to smoke at younger ages, increasing their risk for NCDs (Fig. 77.5). More than 40 million people smoke in Africa. About 1 in 10 adolescents smoke cigarettes and the same proportion uses other tobacco products, such as chewing tobacco, snuff, or pipes.[6] Harmful alcohol consumption, a risk factor for NCDs and injuries, is expected to increase with further economic development.[8] Africa already has the highest prevalence of heavy episodic drinking of any region.

The burden from cancer is expected to more than double between 2008 and 2030, with annual deaths rising from 512,000 to 1.2 million over that period. The drivers for this increase include HIV and other infections, tobacco use, occupational and environmental risks, such as air pollution and exposure in mining, and the aging of the population. The proportion of elderly people is projected to double in many African countries between 2000 and 2030, accounting for a substantial proportion of the projected increase in cancer.[8]

Injuries, such as those related to violence or road traffic accidents, also contribute a substantial disease burden in Africa. Sub-Saharan Africa has the highest road traffic death rate in the world, estimated at 24.1 people per 100,000 population per year.[8,9] Conflicts are another major cause of death and disability.[5] The deadliest of these occurred in Nigeria, where the militant group, Boko Haram, is active. Sudan and South Sudan are also suffering from conflict as are a number of other African countries. Although most of the deaths and injuries affect men, there are high rates of sexual violence against women in conflict situations, as documented for the Liberian civil war and the Rwandan genocide. In conflict-affected countries, poverty levels are usually high and welfare levels low. The stability and social cohesion necessary for development are lacking and there is often breakdown of death registration and other statistical monitoring systems.

Improvements in HIV/acquired immunodeficiency syndrome (AIDS)-related mortality rates and life expectancy have been recorded in many African countries, mainly as a result of the increasing availability of antiretroviral treatments. This major public health achievement is the result of a global response involving enormous commitment and solidarity, strong partnerships, generous funding and other support, and far-sighted innovations.[10] In Southern Africa, which has the highest HIV prevalence, life expectancy at birth, which fell from 62 years in 1990 to 1995 to 53 years in 2000 to 2010, has increased to 59 years in 2010 to 2015.[2]

Major challenges still remain, however, as the coverage and quality of HIV services is insufficient in some large countries with high HIV prevalence. Although declining, the numbers of people acquiring HIV infection are still high. Young people, and especially young women, are most affected. Worldwide, an estimated 250,000 youth aged 15 to 19 years old were newly infected in 2015, with two-thirds living in sub-Saharan Africa.[1] Young women are most vulnerable because of gender-based violence and less access to educational and economic opportunities.

HUMAN RESOURCES FOR NEPHROLOGY

According to the World Health Organization (WHO) Global Health Observatory, over 44% of WHO member states have less than one physician per 1000 population. The African region bears more than 24% of the global burden of disease but has access to only 3% of health workers.[11] The same disparity exists in the nephrology healthcare workforce. Africa has the lowest number of nephrologists per million population (PMP) in the world, and many parts of the continent have no nephrologists at all. A survey conducted by the International Society of Nephrology (ISN) Africa Regional Board in 2014 revealed that, except for North Africa and Mauritius, all African countries have less than two nephrologists PMP, and most have less than one PMP

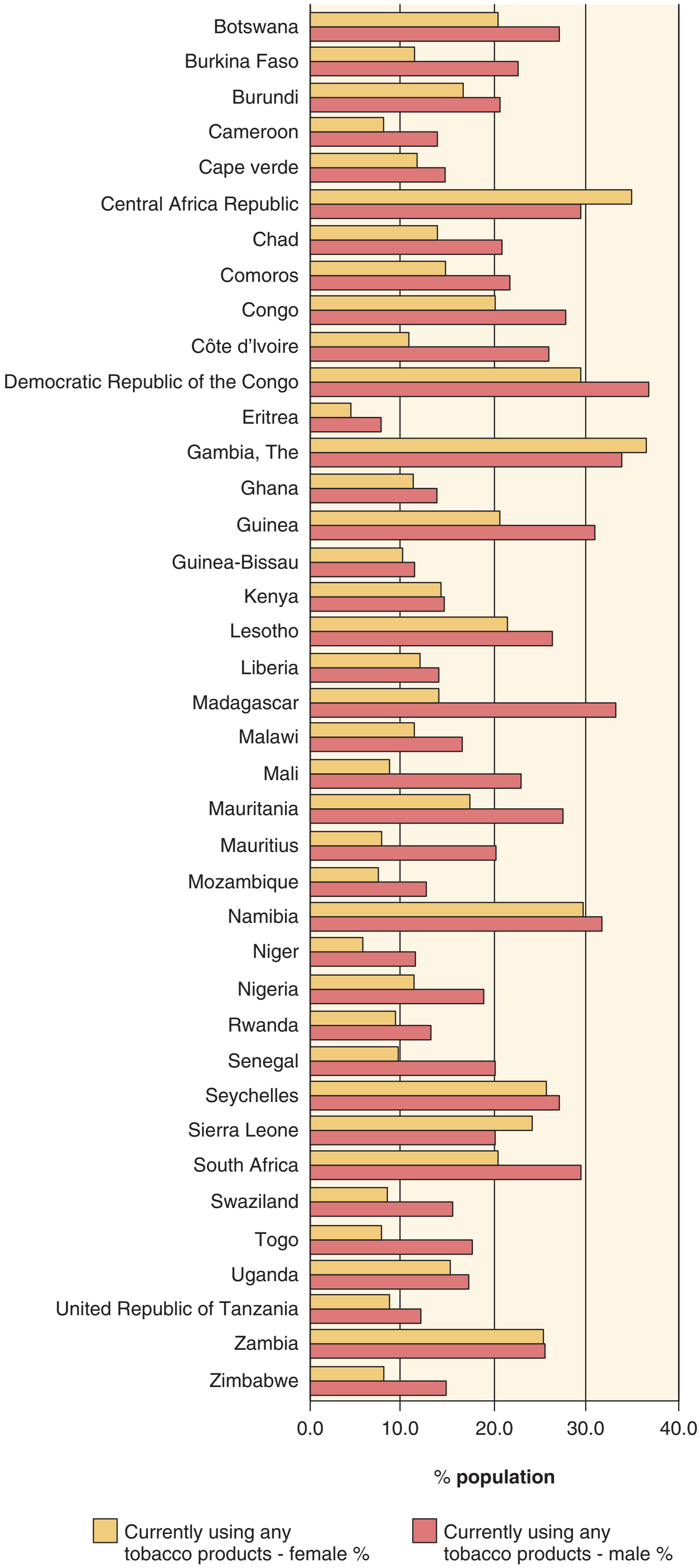

Fig. 77.5 Use of tobacco products among African youth 13–15 years old, 2005–2010. (From Marquez PV, Farrington JL. *The challenge of non-communicable diseases and road traffic injuries in Sub-Saharan Africa: an overview.* Washington, DC: The World Bank; 2013.)

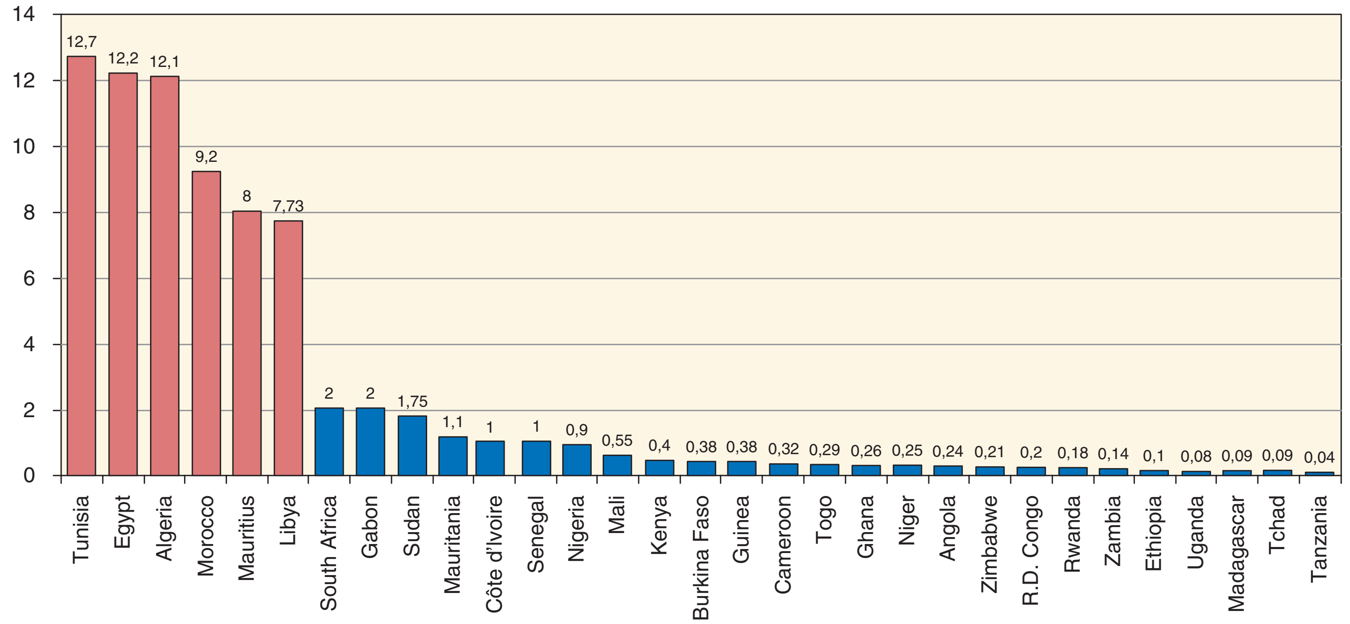

Fig. 77.6 Number of nephrologists per million population in African countries. (From Benghanem Gharbi M. Chronic dialysis programs in African countries: Barriers and successes. World Congress of Nephrology; 2015; Cape Town, South Africa.)

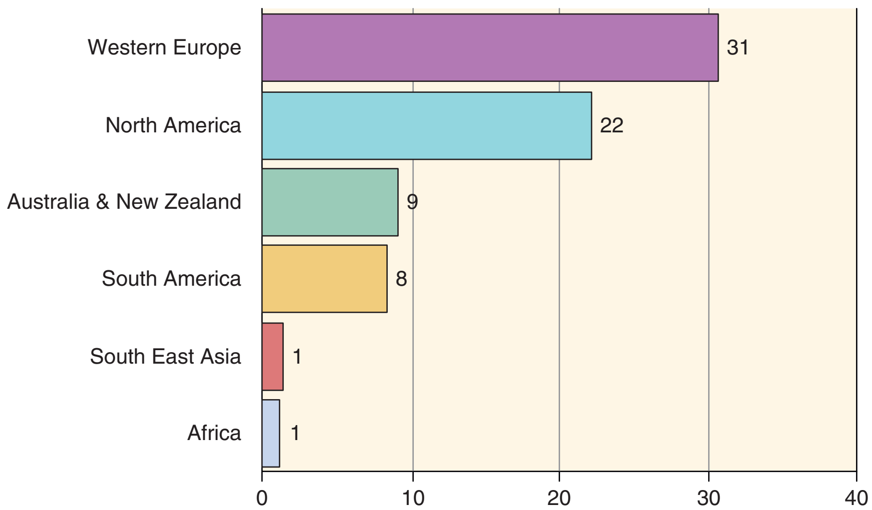

Fig. 77.7 Number of nephrologists per million population, by continent. (From Sharif MU, Elsayed ME, Stack AG. The global nephrology workforce: emerging threats and potential solutions. *Clin Kidney J.* 2016;9:11–22.)

(Fig. 77.6).[12] In contrast, Western Europe and North America have more than 10 nephrologists PMP (Fig. 77.7).[13] In general, nephrologists are clustered around large, densely populated urban centers, which may actually represent optimal use of a scarce resource.[14]

Brain drain continues to pose a great threat to the limited nephrology health workforce on the African continent despite various recommendations[15] and global initiatives.[16] Some of the recommendations proposed include increasing the training of doctors in developed countries, providing assistance to developing countries to aid retention of skilled personnel, restricting the duration of training or work visas for doctors from developing countries, and strengthening specialist training within developing countries.[15]

The ISN Fellowship Program, which started in 1985, is a particularly noteworthy initiative which sponsors physicians from developing countries to undertake nephrology training in centers of the developed world. By the end of 2017, 196 fellows from Africa had been trained, with 84 of them trained in Western European centers, 74 in African centers (mainly in South Africa), and 30 in North America. In a survey conducted in 2010,[16] 90% of ISN fellows indicated that their training had high or good relevance to the needs of their home institution or country, and 85% reported having a high level or good impact on their home institution and home country. Within 10 years of completing their fellowship, 60% of responding fellows occupied a leadership position within their department or hospital, 28% within their country, and 7% at an international level. Fellows who had trained at centers within their region reported fewer difficulties with obtaining visas, more hands-on clinical training, greater relevance and impact on their home units, and less brain drain. However, there were fewer publications reported, and none of high impact.[16]

Because the most effective interventions to reduce the burden of renal disease are likely to be implemented at primary care settings, it would be logical to involve general practitioners and nurses. Providing these categories of healthcare workers with adequate training in the early diagnosis and management of renal disease, with prompt referral when appropriate, could mitigate the effect of low numbers of nephrologists and improve outcomes.[17]

ACUTE KIDNEY INJURY

Epidemiological data for AKI is scarce in Africa. The most comprehensive study is the ISN 0by25 Global Snapshot, conducted over a 10-week period in 2014, to assess the range of AKI seen by physicians in different settings worldwide.[18] African centers contributed 13% of the cases and all the subregions of the continent were represented: North Africa (Egypt, Morocco, and Tunisia), West Africa (Ghana, Niger, Nigeria, and Senegal), Central Africa (Cameroon), East Africa (Ethiopia, Kenya, Sudan, and Tanzania), and Southern Africa (Malawi and South Africa). The mean age of the African patients was 53 years. Most of AKI cases (83.9%) were community-acquired and 19% were seen in intensive care unit (ICU) settings.[19] Dehydration was the most frequent cause of AKI (51.5%), followed by sepsis (32.1%), hypotension (31.8%), and infection (29.8%). The etiologic factors are presented in Fig. 77.8.

It is generally argued that the epidemiology of AKI differs between North Africa and sub-Saharan Africa. In 2016, Olowu et al. published an informative systematic review aiming to assess the outcomes of AKI in sub-Saharan Africa and to identify barriers to care.[20] Among the studies included in this review, 8 were prospective, 32 were retrospective, and one was cross-sectional. AKI was predominantly community-acquired, and the causes are shown in Fig. 77.9. Infection, including malaria, accounted for more than 30% of AKI cases. The other etiologies were dominated by nephrotoxins, pregnancy complications and glomerulonephritis in adults, and by glomerulonephritis, nephrotoxins, hypovolemia, and urinary tract obstruction in children. Most patients presented with severe AKI, with 70% of adults and 66% of children needing dialysis, suggesting that AKI is a more aggressive disorder in sub-Saharan Africa.[20] The likely reasons for this include late presentation to hospital and reliance on clinical criteria for the diagnosis of AKI.

From North Africa, Bourial et al.[21] conducted a 6-month, prospective, multicenter survey (September 2012 to February

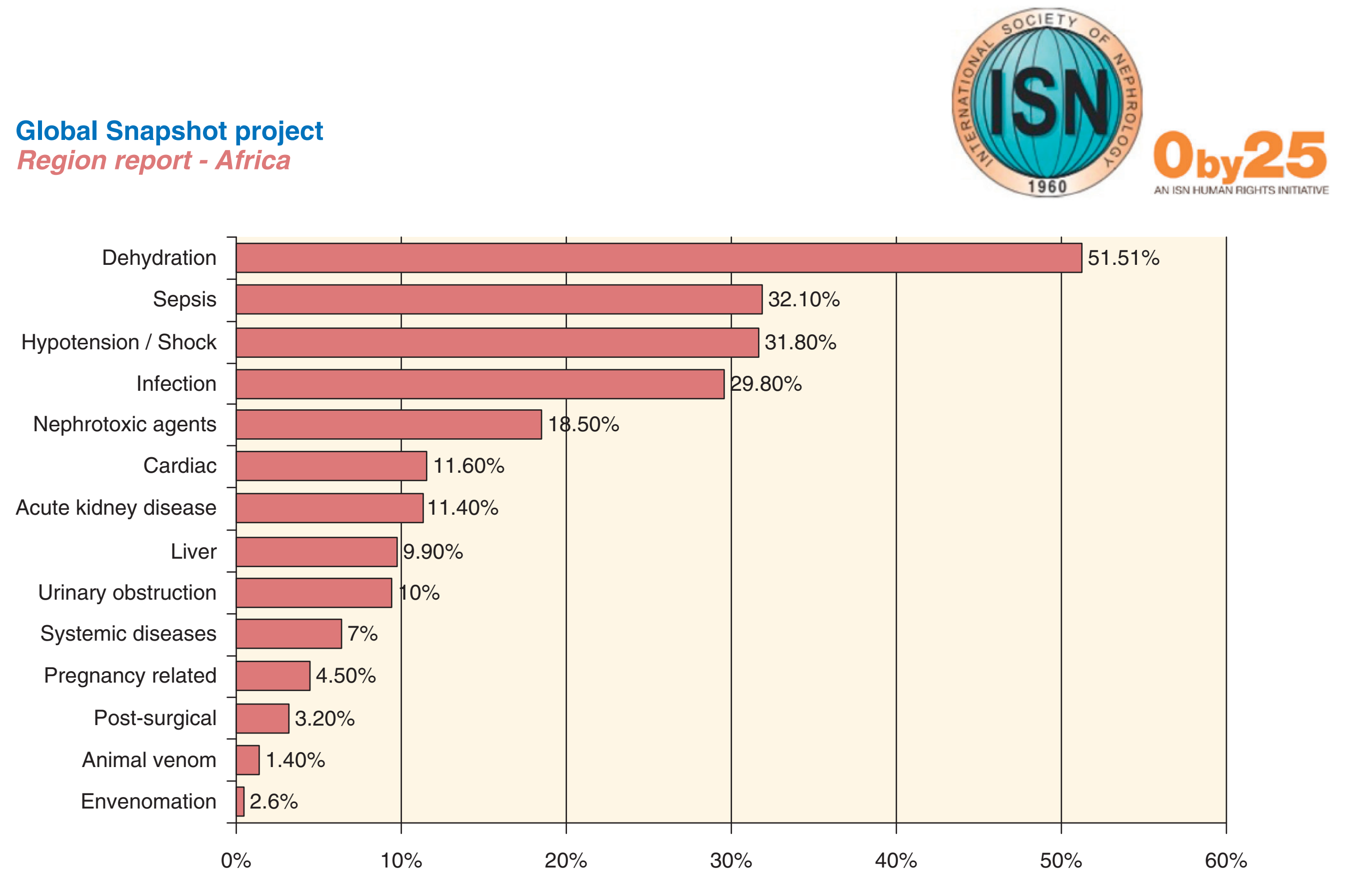

Fig. 77.8 Etiologic factors for acute kidney injury in Africa. (From Mehta RL, Macedo E, Zhang J, ISN 0by25 Initiative Steering Committee. International Society of Nephrology 0by25 Global Snapshot. Regional report for Africa, 2015.)

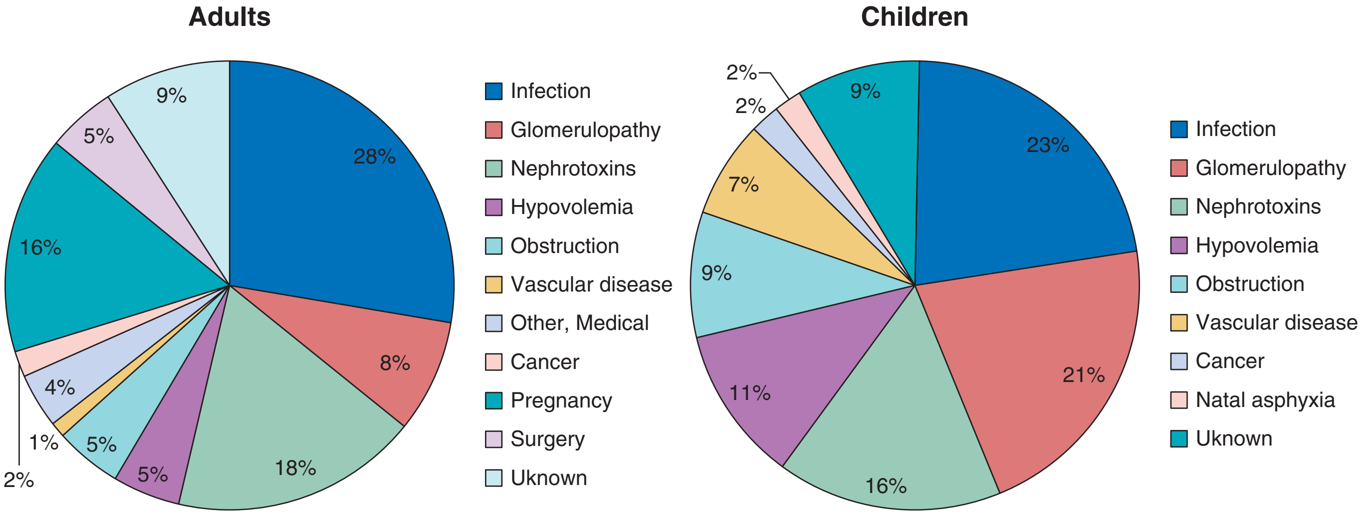

Fig. 77.9 Prevalence of etiologic factors for acute kidney injury in adults and children in sub-Saharan Africa. (Modified from Olowu WA, Niang A, Osafo C, et al. Outcomes of acute kidney injury in children and adults in sub-Saharan Africa: a systematic review. *Lancet Glob Health.* 2016;4:e242-250.)

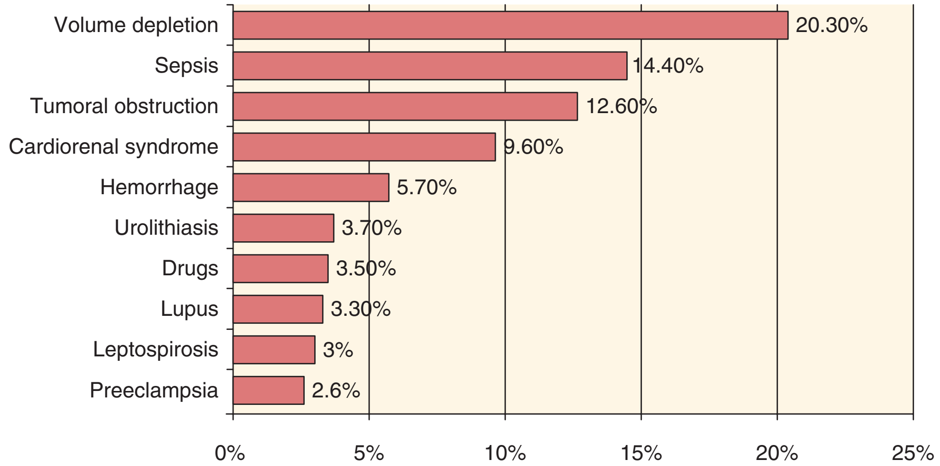

Fig. 77.10 Prevalence of etiologic factors for acute kidney injury in Morocco. (From Bourial M. Aspects epidemiologiques de l'insuffisance renale aigue au Maroc. *Premiere enquete nationale.* Casablanca, Morocco: Université Hassan II de Casablanca; 2014.)

2013) across Morocco (the IRAM Survey) and recruited 541 cases of AKI. Most of the cases (82.4%) were community-acquired and the median age was 57 years (range 3 months–94 years). As shown in Fig. 77.10, volume depletion, sepsis, and tumor obstruction of the urinary tract were the top etiologic factors. Dialysis was required in 40.3% of patients and 21.8% of patients died. Renal function recovered in 25.1% of the cases, whereas 33.6% developed CKD.

Hypovolemia and infection therefore remain the principal causes of AKI in both North Africa and sub-Saharan Africa. Differences in the prevalence of other etiologies are apparent in some regional reports.

ACUTE KIDNEY INJURY IN HUMAN IMMUNIDEFICIENCY VIRUS INFECTION

AKI is a common complication in HIV-infected patients compared with uninfected patients, with an odds ratio of 2.82 for those treated with ART, and 4.62 in the pre-HAART era.[22] Patients hospitalized with complications of HIV may be at increased risk of AKI due to volume depletion, hemodynamic stress, infections, malignancy, and administration of nephrotoxic medication or radiocontrast material.

Kidney injury related to ART occurs in less than 10% of patients. Tenofovir disoproxil fumarate (TDF), a first-line ART agent, is potentially nephrotoxic, as it causes mitochondrial toxicity in proximal renal tubular cells. This may result in AKI, CKD, or Fanconi syndrome. Although TDF is well tolerated by most patients, there is an increased risk of TDF-related renal injury in patients with preexisting renal impairment, older age, low body weight, advanced HIV disease, comorbidities (e.g., diabetes, hypertension and hepatitis C coinfection), concomitant use of other nephrotoxic drugs, and protease inhibitors.[23]

AKI in hospitalized HIV-infected patients is associated with a sixfold higher risk of in-hospital mortality.[23] The most important risk factors include severe levels of immunosuppression (CD4 count <200 cells/mm^3) and opportunistic infection. The most common causes are acute tubular necrosis and thrombotic microangiopathy. After initiation of ART, HIV-infected patients with AKI still have an increased risk of in-hospital mortality, and episodes of AKI seem to be more frequent in the first year of ART. AKI has been associated with infection, hypotension, and exposure to nephrotoxic antimicrobial agents such as aminoglycosides and amphotericin B.[23] Risk factors are low CD4 count, coinfection with hepatitis C, acute or chronic liver injury, diabetes mellitus, and underlying CKD. Glomerular disease in HIV-infected patients (HIV-associated nephropathy [HIVAN], HIV-related immune complex disease [HIV-ICD], and other pathologies)[24–26] can also manifest as AKI and may be the underlying condition upon which an additional insult has precipitated an acute decline in kidney function.

Data on AKI with HIV infection in developing countries are emerging. According to a 2005 to 2006 survey from Johannesburg, South Africa, 101 of 700 patients with kidney failure (14.8%) were HIV-positive, with a mean age of 38 years.[27] Sepsis was the cause of AKI in 60% of the HIV-positive patients. Mortality occurred in 44% of these patients and in 47% of a control group of HIV-negative patients. Hyponatremia, hyperphosphatemia and anemia were predictors of mortality in the HIV-positive patients.[27]

A review of 117 HIV-positive patients with AKI who received acute dialysis in Cape Town, South Africa, revealed that sepsis was present in over 50% at presentation. Acute tubular necrosis was diagnosed in 58%; recovery of renal function occurred in 33.3% while mortality occurred in 41%.[28] A recent study from Bloemfontein, South Africa, reported high mortality among HIV-infected ICU patients needing continuous renal replacement therapy (CRRT). These patients had a mortality rate of 60% as compared with 10% among the HIV-uninfected patients.[29]

ACUTE KIDNEY INJURY AND MALARIA

According to the World Malaria Report, 445,000 deaths were caused by malaria in 2016 and 90% of these deaths occurred in sub-Saharan Africa.[30] Just over half of the people at risk in sub-Saharan Africa were using the primary prevention method, sleeping under an insecticide-treated mosquito net. Although this represents a significant increase, the report also documents a decline in the use of another important preventative measure, spraying the inside walls of homes with insecticides.[30]

Malaria-associated AKI is usually due to *Plasmodium falciparum* infection. The incidence of AKI approaches 60% in patients with a severe form of malaria, with risk factors including severe parasitemia, children under the age of 5 years, pregnant women, and HIV infection.[31] The mortality rate is approximately 45%, and highest in children.[32] Early antimalarial treatment with artemisinin-based therapies, good fluid and electrolyte management, and dialysis when indicated, is associated with improved patient survival and recovery of renal function in patients with malaria-induced AKI. See Chapter 79 for a detailed discussion on malaria and the kidney.

ACUTE KIDNEY INJURY AND TOXINS

The use of traditional medicines is very common in developing countries, where its use is related to spiritual beliefs, the need for protection or cleansing, sexual potency or fertility, suspicion of conventional medicine, and poor access to medical care.[33] In sub-Saharan Africa, more than 80% of people use traditional medicines as their principal form of healthcare. Traditional medicines are used for numerous conditions, including CKD, hypertension, and diabetes, in both urban and rural settings, and by people of all socioeconomic classes (Table 77.3).[34]

These medicines are mostly used without major adverse effects; however, many have been associated with kidney injury. The use of traditional medicines may also delay patients' presentation to medical care for many conditions, including kidney disease.[34] Volume depletion is a major risk factor for the development of AKI associated with traditional medicines. In developing countries, such remedies account for up to 35% of AKI cases and the reported mortality rates range from 24% to 75%.[33] In those who survive, renal recovery is often incomplete, resulting in CKD. As the use of traditional medicines is generally underreported to physicians, it should always be considered in the differential diagnosis of any community-acquired or unexplained renal injury.[35]

Components of traditional medicines that may be associated with AKI include cancer bush (*Sutherlandia frutescens*), ysterhouttoppe (*Dodonaea angustifolia*), marking nut tree (*Semecarpus anacardium*), Bao Gong Teng (*Erycibe obtusifolia Benth*), spearmint (*Mentha spicata*), Opuntia megacantha, and Alder buckthorn bark and berry (*Rhamnus frangula*).[33] The nephrotoxicity of a remedy may also be related to contamination of the remedy, the erroneous identification or preparation of plants, incorrect use, interaction with other medications, and patient factors such as comorbid conditions, age, and sex.[33]

The mechanism of nephrotoxicity is mostly unknown. Examples where the pathophysiology has been identified include aristolochic acid (proximal tubular dysfunction due to inactivation of antioxidant enzymes),[36] atractyloside, the toxic ingredient in impila (inhibits mitochondrial oxidative phosphorylation),[37] and star fruit juice (induces apoptosis in renal tubular cells with precipitation of calcium oxalate crystals).[38] The wide range of renal pathologies reported includes glomerular injury (bladderwrack), Fanconi's syndrome (aristolochic acid), distal renal tubular acidosis (coneflower), nephrogenic diabetes insipidus (bladderwrack), papillary necrosis (willow bark), interstitial nephritis, acute tubular necrosis, kidney stones (djenkol beans), and urothelial malignancies (aristolochic acid). In renal transplant recipients, traditional medicines may also affect allograft function through immune activation (alfalfa and black cohosh) or by influencing the metabolism of immunosuppressive medications (St. John's wort, an inducer of cytochrome P450 3A4). Many electrolyte abnormalities have been associated with the use of traditional medicines, either as a result of volume overload or depletion, or as a result of renal tubular toxicity.[33]

Strategies targeting kidney diseases in LMICs require engagements with key community stakeholders, including practitioners and users of traditional medicines.[33,34] Healthcare practitioners should be proactive in obtaining a history of remedy use, in carefully documenting the clinical features, pathophysiology and renal pathology, and in reporting adverse events so that more comprehensive data can be collected.[35]

Table 77.3 Examples of Traditional Medicines Used Across Sub-Saharan Africa That Have Known Nephrotoxic Effects

Nomenclature			
Scientific Name	**Common English Name**	**Common Uses or Indications**	**Nephrotoxic Effects**
Aloe vera species including ferox and secundiflora	Cape aloes Aloe vera	Southern Africa: arthritis, burns/skin conditions, hypertension, purging/laxative, dyspepsia, antiinflammatory, cosmetics, eye ailments/conjunctivitis, venereal diseases, infertility, impotence East Africa (Kenya, Uganda, Ethiopia, and Tanzania): malaria, purging/laxative for cleansing purposes, dyspepsia, skin ulcerations/wound healing, cosmetic, infertility, antiparasitic	Volume depletion and electrolyte imbalance Acute tubular necrosis Acute interstitial nephritis
Euphorbia matabalensis	Three-forked shrub (Euphorbia)	Eastern Africa (including Somalia, Kenya, Tanzania, Malawi, Mozambique, and Zimbabwe): constipation, purging, hypertension Southern Africa (including Zambia, Botswana, Angola, and Namibia): Abortifacient, constipation, purging, hypertension, venereal diseases, dyspepsia, myalgias	Acute tubular necrosis: directly nephrotoxic, perhaps attributable to latex compounds
Cymbopogon citrullus	Lemongrass	South Africa: diabetes, oral thrush Nigeria: fever/malaria, stimulant, antispasmodic Cameroon: malaria, jaundice Angola; cough, antiemetic, antiseptic, arthritis/arthralgia	Volume depletion/diarrhea Chronic decrease in glomerular filtration rate: chronic interstitial nephritis
Securidaca longepedunculata	African violet tree Wild wisteria	Zambia: abortifacient Burkina Faso: malaria, skin disorders Nigeria: abortifacient, constipation, fever, aphrodisiac Botswana and Eastern Africa (including Tanzania, Malawi, Uganda, Kenya, and Zimbabwe): epilepsy, arthritis, purging, constipation, wound healing, snakebites, fever/malaria, impotence, dysmenorrhea	Acute tubular necrosis: causes cortical necrosis, volume depletion, and contains salicylates thought to contribute to renal vasoconstriction Acute interstitial nephritis
Crotalaria laburnifolia	Wild sunhemp Rattlepod Birdflower	South Africa, Botswana, and Eastern Africa (including Mozambique and Zimbabwe): enema, purging, dysmenoritiea, abortifacient, dyspepsia, antispasmodic	Acute tubular necrosis: contains nephrotoxic alkaloids Hepatorenal failure
Callilepsis laureola	Ox-eye daisy Impila	Mozambique and South Africa: dyspepsia, antiparasitic, impotence/infertility, purging, evil spirits	Acute tubular necrosis and acute interstitial nephritis Hepatorenal failure and chronic decrease in glomerular filtration rate Hyperkalemia

From Stanifer JW, Kilonzo K, Wang D, et al. Traditional medicines and kidney disease in low-and middle-income countries: opportunities and challenges. *Sem Nephrol*. 2017;37:245-259.

Other nephrotoxic agents encountered in some African countries, especially in East Africa and the Maghreb, include hair dye preparations that contain paraphenylenediamine.[39] Some of these dyes are added to henna, a traditional cosmetic agent applied to the upper and lower limbs, for its skin staining effect. Paraphenylenediamine can be absorbed through the skin, but more severe intoxication results from ingestion of the hair dye, mostly in suicide attempts.[40] AKI is mainly due to acute tubular necrosis caused by rhabdomyolysis. Mortality may result from other manifestations of the poisoning—for example, angioedema, which occurs early, and arrhythmias caused by direct cardiotoxicity of the chemical.[39] There is no known antidote.

Animal toxins are also a major cause of AKI in Africa. Snake venoms are a complex mix of different toxic and nontoxic proteins and peptides, nonprotein toxins, carbohydrates, lipids, amines, and other small molecules. The toxins of most importance in human envenoming include those that affect the nervous, cardiovascular, and hemostatic systems, and cause tissue necrosis.[41]

AKI developing after snake bites may be related to intravascular hemolysis, hypotension, blood hyperviscosity, myoglobinuria, and hemorrhage. Bites occur especially in plantations and commonly involve *Causus maculatus* (spotted night adder), *Naja melanoleuca* (black cobras), and *Dendroaspis* species (green mambas). In forested areas, gaboon vipers (bitis species) are common while in savanna areas *Echis* species are common and cause the greatest numbers of deaths by envenomation in Africa.[42] The range of renal pathology includes acute tubular necrosis, cortical necrosis, acute interstitial nephritis, and proliferative glomerulonephritis.

Scorpion stings are a significant public health problem in north-Saharan Africa, Sahelian Africa, and South Africa.[43] The scorpion α-toxins bind to voltage-gated sodium channels to cause neuronal excitation.[44] AKI may develop in some patients as a result of cardiac dysfunction, disseminated intravascular coagulation and hemorrhage.

PREGNANCY-RELATED ACUTE KIDNEY INJURY

Obstetric AKI has become rare in industrialized countries, whereas it continues to be a frequent clinical problem in many African countries, mainly secondary to preeclampsia and eclampsia, septic conditions, and obstetric hemorrhage (see also Chapter 48). In Morocco, pregnancy-related AKI occurred mostly in the third trimester (66.6% of affected patients) and in the postpartum period (25%). Preeclampsia and hemorrhagic shock (25%) were reported as the major causes of AKI.[45] In a retrospective study of patients admitted to an ICU for eclampsia in Nigeria from 2001 to 2005, Okafor and Efetie[46] found a case-fatality rate of 29%, with 9.1% of deaths related to AKI. The incidence of eclampsia was inversely related to good antenatal care, standard of living, and literacy rate. In a prospective study of 178 consecutive women with eclampsia in Morocco, Mjahed et al.[47] found that the incidence of AKI was 25.8%. Dialysis was needed in one-third of the patients, and mortality rates were higher among patients with AKI than those without AKI (32.6% vs. 9.1%).

THERAPEUTIC CHALLENGES

In most African countries, low availability of resources and inadequate health infrastructure are associated with poor recognition and treatment of AKI. As illustrated in the systematic review of Olowu et al., AKI in sub-Saharan Africa is severe, mainly because of the late presentation of patients to tertiary care centers.[20] In the studies reporting patients who needed dialysis, 66% of children and 70% of adults had indications for dialysis. However, only 64% of children (across 11 studies) and 33% of adults (four studies) received dialysis when needed. The overall mortality was 34% in children and 32% in adults but rose to 73% in children and 86% in adults when dialysis was needed but not received. Major barriers to access to care were unaffordable out-of-pocket costs, limited hospital resources, late presentation, and female sex.[20] In another review of AKI publications on Nigerian children between 1990 and 2012, dialysis access rates ranged between 15.2% and 68.2% of those with a medical indication for dialysis.[48]

This contrasts with the practice in North Africa, where dialysis for AKI is more easily accessible in hospital settings. In the Moroccan survey, dialysis was needed in 40.3% of cases and intermittent hemodialysis was the predominant prescribed modality (96%). At discharge, the mortality was 25.4%, dialysis dependence was 6.8%, residual renal failure was present in 39.2% and there was complete renal recovery in 29.3%.[21]

Two recent global initiatives have the potential to have a major impact on the outcomes of patients with AKI, especially in low-resource settings. The ISN has launched “0by25”, an ambitious agenda to eliminate all preventable deaths from AKI by 2025.[49] The 0by25 project will focus on evidence and awareness, and on education and training for health workers to increase the early detection of AKI and the appropriate use of simple management, such as fluid repletion.

The second global initiative is “Saving Young Lives”, which is focused on AKI in children and aims to develop sustainable acute peritoneal dialysis (PD) programs in very low-resource settings.[50,51] PD is now being increasingly utilized to manage patients with AKI and can be life-saving in situations where hemodialysis is not available. Outcomes appear to be similar when compared with treatment with extracorporeal therapies.[52,53] Support provided by organizations such as the International Society for Peritoneal Dialysis (ISPD), the International Pediatric Nephrology Association (IPNA), the ISN and the Sustainable Kidney Care Foundation (SKCF) have allowed physicians working in resource-constrained regions of Africa to develop expertise in the provision of acute PD. Practical hands-on training has already been provided to more than 50 nurses and physicians at an annual training course held at the Red Cross War Memorial Children’s Hospital in Cape Town, South Africa. This course covers all aspects of acute PD, including catheter insertion, fluid prescription, and clinical problem solving.

It is important to note that a large proportion of AKI cases and their clinical consequences are potentially preventable.[54] Education regarding the avoidance of nephrotoxins, prompt treatment of infections, and fluid replacement are therefore important facets of therapy.[55] This approach is vital because many countries in Africa are still not able to provide renal replacement therapies.

CHRONIC KIDNEY DISEASE

Chronic kidney disease (CKD) is a major public health problem, with an estimated global prevalence of 11% to 13%.[56] In Africa, the increases in NCDs, pregnancy-related disorders and injuries, and the high burden of infectious diseases may all contribute to the epidemic of CKD. People from disadvantaged communities are at particularly high risk of unrecognized and untreated CKD.[57] This includes indigenous communities, and racial and ethnic minorities.[58]

CKD is defined as the presence of abnormalities of kidney structure or function, with implications for health, for more than three months.[59] In practice, this is usually determined by finding a glomerular filtration rate (GFR) of below 60 mL/min/1.73 m^2 or markers of kidney damage such as proteinuria, hematuria, or abnormalities on renal imaging. The latest KDIGO guidelines[59] recommend that CKD classification be based on cause, GFR category, and degree of albuminuria. GFR should be estimated using serum creatinine and a creatinine-based prediction equation.[59]

ISSUES IN THE DIAGNOSIS OF CHRONIC KIDNEY DISEASE

The accurate diagnosis and staging of CKD are especially important in Africa, as options for treating late complications such as ESRD are very limited. The screening of those at higher risk of developing CKD is therefore essential. The choice of methods used to assess GFR can have a major impact on determining if an individual has CKD or not, on the stage of CKD, and on the reported population prevalence.

Different assays may result in interlaboratory differences in serum creatinine and therefore estimated GFR.[60] All creatinine assays should therefore be standardized by calibration to isotope-dilution mass spectrometry (IDMS)-traceable reference material. Enzymatic methods are preferred as they have greater analytical precision than the colorimetric Jaffe reaction-based methods.

GFR prediction equations need to be validated in local populations against gold standard methods. Three African studies, two from South Africa and one from Côte d'Ivoire, have compared prediction equations with GFR measured by gold standard methods. Van Deventer et al.[61] studied 100 black South Africans and measured GFR using the plasma clearance of chromium-51-EDTA (^{51}Cr-EDTA). Serum creatinine was measured using an IDMS-traceable assay. Madala et al.[62] studied 91 participants of African and 57 of Indian origin, and measured GFR using technetium-99m-diethylenetriaminepentaacetic acid (^{99m}Tc-DTPA). Both studies reported a good performance of the 4-variable MDRD equation in black South Africans, and both concluded that the correction factor for black ethnicity should not be used. Madala et al. suggested that a new correction factor, or a new equation, is needed for South Africans of Indian origin. Sagou Yayo et al.[63] used iohexol to measure GFR in 120 blood donors in Abidjan and also found that the performance of the CKD-EPI and MDRD prediction equations was better without the "African-American" ethnicity factor.

International guidelines recommend the use of the Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI) prediction equation, unless an alternative equation has been shown to improve the accuracy of GFR estimates.[59,64] This equation is considered more accurate than the previously recommended Modification of Diet in Renal Disease (MDRD) study equation, which systematically underestimates GFR at higher values.

Although current guidelines use a universal GFR threshold of 60 mL/min/1.73 m^2 for the diagnosis of CKD, this may not be appropriate in all cases. Pottel et al.[65] have suggested that a cut-off of 75 mL/min/1.73 m^2 be used for children and young adults because they may have low GFR values, below the third percentile for age and sex, but still above 60 mL/min/1.73 m^2. At the other end of the age spectrum, there have been calls for caution in labeling older patients as having CKD if they have mild, stable reductions in GFR (between 45 and 60 mL/min/1.73 m^2) without any other evidence of kidney damage or systemic disease.[66]

Many cross-sectional epidemiologic studies suffer from a common weakness in that they do not include repeat estimates of GFR or protein excretion and hence do not meet the criterion of chronicity for diagnosing CKD. People with abnormal kidney function or damage are inferred to have CKD if there is not an acute illness likely to cause acute kidney injury. When repeat measurements are done, however, this will often result in lower estimates for CKD prevalence.[67]

CHRONIC KIDNEY DISEASE PREVALENCE IN AFRICAN COUNTRIES

The systematic review on the population prevalence of CKD in sub-Saharan Africa by Stanifer et al.[68] identified only 18 medium-quality studies and three high-quality studies for inclusion in their meta-analysis. These studies reported data from 13 African countries (Fig. 77.11). The authors reported an overall prevalence of 13.9% but highlighted the need for more studies of good quality. Country estimates ranged from 2% in Cote d'Ivoire to 30% in Zimbabwe. Nearly half of the countries represented had prevalence estimates ranging between 4% and 16%.

More recently, population prevalence studies have been published from a number of African countries. From Cape Town, South Africa, Matsha et al.[69] reported a crude prevalence of 17.3% in a geographical cohort, while Adeniyi et al.[70] reported an age-adjusted prevalence of 6.4% in a cohort of teachers. The higher estimates by Matsha et al. may at least partly be explained by the very high burden of hypertension, diabetes, obesity, and smoking in their cohort. In northern Tanzania, the CKD AFRiKA Study reported a prevalence of 7.0%, with a much higher prevalence in urban than in rural areas,[71] whereas Seck et al. reported a prevalence of 4.9% in a population-based survey in northern Senegal.[72] Kaze et al. found a CKD prevalence of 10% in an urban district of Cameroon[73] and Kalyesaluba et al. reported a prevalence of 9.8% in central Uganda.[74] In Nigeria, Wokoma et al.[75] summarized the results of 30 community-based CKD screening studies, which included more than 17,000 participants, and found an overall crude CKD prevalence of 11.7%. The prevalence was much higher in rural than in urban communities (16.4% vs. 7.0%).

The large Maremar study[67] included over 10,000 Moroccan adults and reported a population prevalence of 5.1%. This study by Benghanem Gharbi et al. is especially noteworthy in that it is one of the few studies of CKD prevalence in which particular care was taken to satisfy the criterion of chronicity in diagnosing CKD.[67,76]

Kayange et al. have published a systematic review of community-based studies describing the prevalence of renal abnormalities among children in sub-Saharan Africa.[77] Thirty-two studies from nine countries were included and these reported widely varying results. The mean prevalence of proteinuria was 32.5% (2.2%–56.0%), hematuria 31.1% (0.6%–67%), and imaging abnormalities 14.8% (0.5%–38.0%). The quality of these studies was graded as moderate to poor. Few were recent, and none included the calculation of eGFR. Most of the studies were performed in areas where schistosomiasis is endemic, and this was the main risk factor identified.

COMMON CAUSES OF CHRONIC KIDNEY DISEASE AND END-STAGE RENAL DISEASE

In many cases where CKD or ESRD is identified in African patients, a specific diagnosis is never established. This may be because renal biopsy services are often not available or

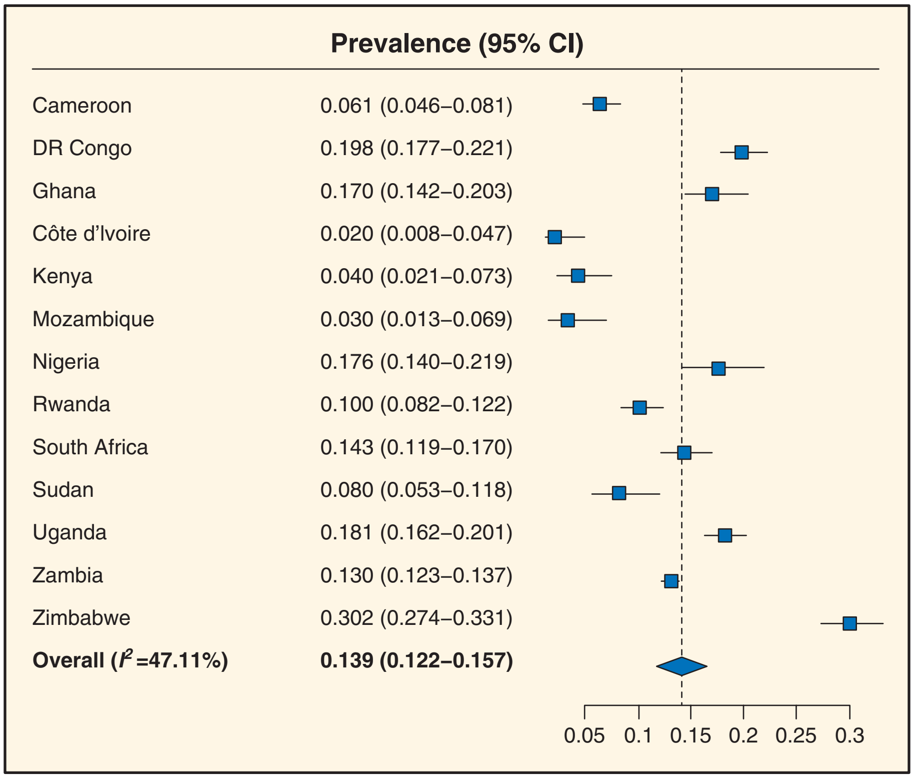

Fig. 77.11 Forest plot of meta-analyses showing the pooled prevalence of chronic kidney disease by country. (From Stanifer JW, Jing B, Tolan S, et al. The epidemiology of chronic kidney disease in sub-Saharan Africa: a systematic review and meta-analysis. *Lancet Glob Health.* 2014;2:e174-e181.)

Table 77.4 Etiology of End-Stage Renal Disease (% of Total) Reported in Registry Data From African Countries

Study	Country	Last Data	Cause Unknown	Hypertension	Diabetes	GN	Cystic Disease
Afifi, 2008[83]	Egypt	2008	15.2	36.6	13.5	8.7	3.2
Alashek, 2012[84]	Libya	2010	7.3	14.6	26.5	21.2	6.3
Albitar, 1998[85]	Reunion Island	1996		27.5	33.6	13.1	5.8
Davids, 2017[79]	South Africa	2015	34.1	33.7	14.4	9.5	2.9
Elamin, 2010[80]	Sudan	2009	42.6	26.1	10.4	5.5	2.6
Tunisian Ministry of Health, 2011[81]	Tunisia	2010	29.2	13.6	21.0	13.0[86]	3.4
Moroccan Society of Nephrology and Ministry of Health, 2013[82]	Morocco	2007	37.9	10.1	17.8	8.5	3.4

GN, Glomerulonephritis.

because patients present late, with poor renal function, hypertension and shrunken kidneys. A presumptive diagnosis of chronic glomerulonephritis or of hypertensive renal disease is often made, without good grounds for making such a diagnosis.[78] It has been recommended that the label "CKD/ESRD etiology uncertain/unknown" be used in such cases, and whenever there is uncertainty about the renal diagnosis.

Data on the etiology of CKD/ESKD is available from renal registries for some African countries (Table 77.4). The "CKD/ESRD unknown" category was indicated as the most common renal diagnosis in reports from South Africa,[79] Sudan,[80] Tunisia,[81] and Morocco.[82]

DIABETIC NEPHROPATHY

The International Diabetes Federation has estimated that three-quarters of people with diabetes live in low- or middle-income countries and that the number of adults with diabetes will increase from 415 million in 2015 to 642 million in 2040.[87] The Africa region is projected to experience the highest growth rates in the number of diabetic patients, with an increase of 140.7% by 2040. These increases are driven

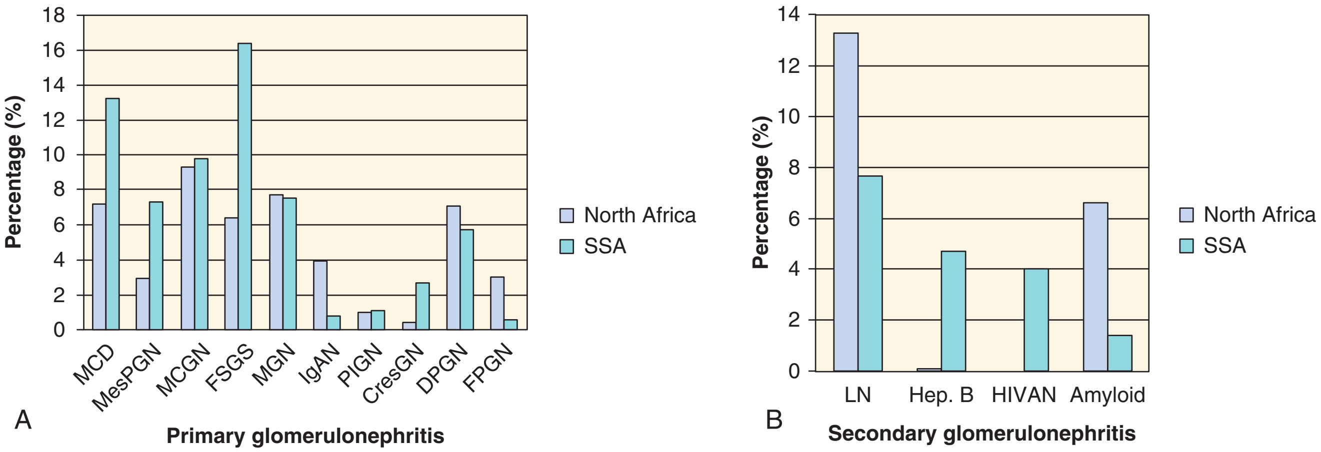

Fig. 77.12 Differences between North Africa and sub-Saharan Africa in the prevalence of common primary (A) and secondary (B) glomerular diseases. *MCD*, Minimal change disease; *MesPGN*, mesangioproliferative GN; *MCGN*, mesangiocapillary GN; *FSGS*, focal segmental glomerulosclerosis; *MGN*, membranous GN; *IgAN*, IgA nephropathy; *PIGN*, postinfectious GN; *CresGN*, crescentic GN; *DPGN*, diffuse proliferative GN; *FPGN*, focal proliferative GN; *LN*, lupus nephritis; *Hep B*, hepatitis B virus-associated nephropathy; *HIVAN*, HIV-associated nephropathy. (From Okpechi IG, Ameh OI, Bello AK, Ronco P, Swanepoel CR, Kengne AP. Epidemiology of histologically proven glomerulonephritis in Africa: a systematic review and meta-analysis. *PLoS ONE.* 2016;11:e0152203.)

by sedentary lifestyles, unhealthy diets, obesity, population growth, and aging.

The rates of diabetes-related complications, including diabetic nephropathy, have slowed markedly over the past two decades, as a result of improved medical care for blood pressure, glycemic control, lipid management, and the use of agents that block the renin-angiotensin system. For example, in the US, the rate of patients with diabetes reaching ESRD declined by 28% between 1990 and 2010.[88] However, a large burden of disease persists because of the continued increase in the prevalence of diabetes.

Noubiap et al.[89] have published a systematic review of diabetic nephropathy in Africa that included 32 studies from 16 countries. They found that measurement of urine protein excretion was the most common method of assessing kidney damage in diabetic patients and that the reported overall prevalence of CKD in diabetics varied from 11% to 83.7%. The rate of ESRD at 5 years of follow-up was 34.7% and the mortality rate from nephropathy was 18.4% at 20 years of follow-up. Duration of diabetes, high blood pressure, advancing age, obesity, and glucose control were the common determinants of kidney disease.

GLOMERULONEPHRITIS

Okpechi et al.[90] have conducted a systematic review of histologically-proven glomerulonephritis in Africa, which covered the period from 1980 to 2014. They noted declining renal biopsy rates and the consequent paucity of data on pathologic patterns of renal diseases from Africa. Their review included 24 studies (12,093 biopsies) from 13 countries, with 70% of the studies originating from North African countries. Nephrotic syndrome was the most common indication for renal biopsy. The frequency of common primary renal diseases reported included minimal change disease (MCD) at 16.5%, focal segmental glomerulosclerosis (FSGS) at 15.9%, mesangiocapillary GN (MCGN) at 11.8%, IgA nephropathy at 2.8%, and crescentic GN at 2.0%. Nephritis related to hepatitis B (8.4%) and systemic lupus erythematosus (7.7%) had the highest prevalence among the secondary glomerular diseases. Fig. 77.12 illustrates some of the differences between North African and sub-Saharan African countries.[90]

It is unlikely that quartan malaria due to *Plasmodium malariae* infection causes a specific glomerulopathy which presents as nephrotic syndrome in African children. Almost all of the reports of "malarial glomerulopathy" or "tropical glomerulopathy" were published before 1975; they rest mainly on circumstantial evidence and there seems to be no recent data to support the existence of this entity.[91]

KIDNEY DISEASE ASSOCIATED WITH HUMAN IMMUNODEFICIENCY VIRUS INFECTION

A broad spectrum of renal diseases is seen in patients with HIV infection. This includes diseases directly related to the HIV infection, those related to superinfections, those related to the immune response, and those related to the treatment. HIV-associated nephropathy (HIVAN) is the classic lesion described, presenting with nephrotic-range proteinuria and decreased renal function but, because of renal salt wasting, patients often do not have hypertension or significant edema. Renal histology reveals a collapsing form of FSGS, with associated cystic tubular dilatation, interstitial inflammation, and the presence of tubuloreticular inclusion bodies. HIV RNA has been demonstrated in podocytes and tubular epithelial cells.[92]

HIVAN was first reported in 1984,[93–95] with the first reports from Africa published in the late 1980s and early 1990s.[96–98] The worldwide prevalence is highest in sub-Saharan Africa,[99] occurring especially in the absence of antiretroviral therapy and in patients with markedly reduced CD4 counts and elevated viral loads.[100]

Individuals of African descent have a genetic susceptibility to the development of HIVAN that is related to single nucleotide polymorphisms in the apolipoprotein L1 (APOL1) gene.[101] The G1 variant includes two missense mutations

(S342G, Rs73885319 and I384M, Rs60910145) that are in close linkage disequilibrium, and the G2 variant is a two-base pair deletion (N388del, Y389del, Rs71785313). These variants confer risk for HIVAN and HIV-associated FSGS, as well as other glomerular diseases and hypertensive nephrosclerosis. Kasembeli et al.[102] have demonstrated that HIV-positive, ART-naïve black South Africans with two APOL1 risk alleles are at very high risk of developing HIV-associated nephropathy. In their study, 79% of patients with HIV-associated nephropathy and only 2% of controls carried two risk alleles. Individuals carrying any combination of two risk alleles had 89-fold higher odds of developing HIV-associated nephropathy compared with HIV-positive controls. It is estimated that 50% of HIV-positive individuals who have two APOL1 risk alleles and do not receive effective ART will develop HIVAN.[103]

There is a wide variability in the prevalence of HIVAN in different African populations and this is likely related to the frequency of the APOL1 risk alleles. The highest frequency of the G1 and G2 variants is found in West Africa whereas populations in North and East Africa have a much lower frequency of the risk alleles and of HIV-related renal disease.[104] For example, in Kenya, Koech et al.[105] performed renal biopsies on ART-naïve, HIV-positive adults who had persistent proteinuria. None of the biopsies revealed HIVAN; acute interstitial nephritis was the most common histological diagnosis in this group of patients.

Renal biopsy data on African patients with HIV infection has mostly come from South African studies. In Durban, Han et al. biopsied 30 HIV-positive patients with proteinuria who were not on antiretroviral therapy and found that 83% had HIVAN. Gerntholtz et al.[24] described the findings in 99 patients from the Johannesburg area: 27% of the biopsies showed classic HIVAN, 21% had HIV-related immune complex disease (HIV-ICD), and more than half had diagnoses which were not directly related to the HIV infection. In those patients with immune complex disease, subepithelial deposits were sometimes associated with a "ball-in-cup" basement membrane reaction pattern. Also, from Johannesburg, the biopsy series of Vermeulen et al.[25] included 364 HIV-infected patients; HIVAN was present in 32.7% and HIV-ICD in 11.8% of these patients. Wearne et al.[26] reviewed the biopsies of 192 HIV-infected patients in Cape Town, reporting that HIVAN was the most common histological diagnosis (57%), with the collapsing subtype of FSGS present in 43%, and describing a "fetal variant" of FSGS in some cases. Immune complex disease was present in 8.3% and a combination of HIVAN and HIV-ICD in 21.9%.

In the absence of antiretroviral therapy, HIVAN progresses rapidly to ESRD. In many countries, the introduction of ART has led to substantial reductions in the incidence of HIVAN and HIVAN-related ESRD.[106] However, there have been conflicting reports on the benefit of ART in patients with HIV-ICD, with a study by Szczech et al.[107] finding that renal lesions other than HIVAN did not benefit from ART. Several African studies have demonstrated improvements in kidney function in patients who have initiated ART for HIV-related renal disease. These include the DART study in Uganda and Zimbabwe,[108] the study by Peters et al. in Uganda,[109] the study by Mpondo et al. in Tanzania,[110] and the South African studies by Wearne et al.[26] in Cape Town and Fabian et al.[111] in Johannesburg. The South African studies included renal biopsy

Table 77.5 Regional Access to Antiretroviral Therapy in 2015 for Africans Living With Human Immunodeficiency Virus. Figures Are Percentages With Confidence Intervals

	Adults (Aged 15 Years+)	Children (Aged 0–14 Years)	Pregnant Women[a]
Eastern and Southern Africa	53% (50–57)	63% (56–71)	90% (82–95+)
Western and Central Africa	29% (24–35)	20% (16–25)	48% (40–58)
Middle East and North Africa	16% (12–24)	20% (16–25)	12% (9–18)
Global averages	46% (43–50)	49% (42–55)	77% (69–86)[114]

[a]Treatment to prevent mother-to-child transmission of human immunodeficiency virus.

From AIDS by the numbers. Geneva, Switzerland: Joint United Nations programme on HIV/AIDS 2016.

data and, unlike the Szczech et al. study,[107] demonstrated benefit in both HIVAN and HIV-ICD.

Globally, there has been a rapid scale-up of antiretroviral therapy. Gains have been greatest in Eastern and Southern Africa, the world's most affected region.[112] South Africa alone had nearly 3.4 million people on treatment in 2015, more than any other country in the world. The latest South African guidelines recommend initiating ART in all HIV-infected persons, irrespective of CD4 count or viral load.[113] Kenya has the next-largest program in Africa, with nearly 900,000 people on treatment. Botswana, Eritrea, Kenya, Malawi, Mozambique, Rwanda, South Africa, Swaziland, Uganda, the United Republic of Tanzania, Zambia, and Zimbabwe all increased treatment coverage by more than 25% between 2010 and 2015.[112] Table 77.5 summarizes the data from the different African regions regarding access to ART.[114] Western and Central African countries have the lowest access, with only 29% of adults, 20% of children, and 48% of pregnant women living with HIV accessing treatment. The global averages are 46%, 49%, and 77%, respectively.

GLOMERULAR DISEASE ASSOCIATED WITH HEPATITIS B VIRUS INFECTION

Hepatitis B virus (HBV) infection is highly endemic with at least 65 million chronic HB surface antigen (HBsAg) carriers in Africa. The major transmission route is horizontal transmission between children under the age of 5 years, and sexual transmission in adolescents and young adults. Healthcare workers are also at risk for parenteral/percutaneous transmission during occupational exposures.[115]

The first case of HBV-associated membranous glomerulonephritis (MGN) was reported in 1971,[116] and the frequency of reported cases reached its peak around 1988–1994. Since then, there has been a decline in the numbers of new cases, largely linked to increasing HBV immunization.[115] Despite the high burden of HBV infection in Africa, there are relatively few reported case series of HBV-associated renal disease, and these are mainly from Southern Africa.[117–123] This is likely related to the availability of renal biopsy and pathology services, including access to electron microscopy.

In one of the larger African case series, Bates et al.[123] studied 71 South African and Namibian children with HBV-associated MGN and compared them with 12 adults who had HBV-associated MGN and 33 adults with idiopathic MGN. The children with HBV-associated MGN more frequently had prominent hematuria, raised serum transaminases, or low serum C3 and C4 levels. Their renal biopsies frequently revealed features of mesangiocapillary glomerulonephritis (MCGN) in addition to the typical subepithelial deposits of MGN. Virus-like bodies and tubuloreticular inclusion bodies were seen in more than 80% of the biopsies and HBeAg was identified in the subepithelial deposits. The remission rate in this cohort of children was 57% at four years; age of 6 years and above at presentation and prominent mesangial deposits on biopsy was associated with fewer remissions and worse renal outcome.

Bhimma et al.[122] documented a significant change in the incidence of HBV-associated MGN over the 6 years following the introduction of the HBV vaccine into the routine childhood immunization schedule in South Africa in 1995. By 2000 to 2001, the incidence of HBV-associated MGN had declined to 12% of the preimmunization rate. The latest data from the South African Renal Registry indicates that 1.3% of all patients on RRT are hepatitis B positive.[79] The prevalence is somewhat lower in public sector treatment centers (1.1% vs. 1.5%) than in private sector centers, where many patients are excluded from RRT because HBV infections are associated with adverse renal transplant outcomes and there is limited access to nucleoside/nucleotide analogues.[124]

POSTINFECTIOUS GLOMERULONEPHRITIS

Poststreptococcal glomerulonephritis follows upper respiratory tract infections or impetigo caused by nephritogenic strains of group A beta-hemolytic streptococcus. It is currently a childhood disease of the developing world, where the burden ranges between 9.5 to 28.5 new cases per 100,000 individuals per year.[125] The condition has become rare in developed nations, where it is now seen mainly in patients over the age of 60 years, especially in association with conditions of immune system compromise such as malignancy, diabetes, alcoholism, or intravenous drug use.[125,126]

Okpechi et al.[90] reported a prevalence of postinfectious glomerulonephritis (PIGN) of 3.5% in their systematic review of biopsy-proven glomerular disease in Africa. The region with the highest reported frequency was North Africa, at 10.3%. In the study by Elzouki et al.[127] poststreptococcal glomerulonephritis was the most common renal disease seen in children admitted to a large referral hospital in Libya. Preceding streptococcal throat infections (occurring in 80%) were much more common than skin infections. In a 30-year review of histologically-proven glomerulonephritis in Tunisia, Ben Maïz et al.[128] reported that the proportion of glomerulonephritis cases with proliferative endocapillary and mesangiocapillary GN fell between the periods 1975 to 1985 and 1995 to 2005, from 15.9% and 21.6% to 6.9% and 7.7%, respectively. This change matched a drop in the incidence of acute rheumatic fever in the Tunisian population, most likely resulting from public health interventions and the widespread use of antibiotics.

The prognosis for PIGN is generally excellent, especially in children, but may be less benign in some cases where hypertension, proteinuria, and loss of renal function may persist after the initial illness or develop some years later.[129] Nasr et al.[126] reported 109 cases in patients older than 65 years, finding a male predominance, and an immunocompromised background in 61%, most commonly diabetes or malignancy. The most common site of infection was the skin, and the most common causative agent was staphylococcus (46%), followed by streptococcus (16%) and unusual gram-negative organisms. Fewer than one-quarter of the patients had full recovery of renal function.

LUPUS NEPHRITIS

Lupus nephritis is a significant cause of mortality and morbidity in patients with systemic lupus erythematosus (SLE) and has been reported to be more aggressive in African populations. The review by Okpechi et al.[90] found lupus nephritis to have a high prevalence among the secondary glomerular diseases (7.7% of all biopsies), with the highest regional prevalence reported from North Africa (13.9%).

The management of African patients with SLE is hampered by the limited availability and high costs of renal biopsy services, immunosuppressive drugs, and laboratory facilities.[130] Ameh et al.[131] conducted a systematic review of the treatment and outcomes of lupus nephritis in Africa, including sixteen studies in their review, with one-half of these being from North Africa. All the studies reported the use of corticosteroids for induction therapy and 15/16 also used cyclophosphamide. Evidence from the Aspreva Lupus Management Study (ALMS)[132] supports the use of mycophenolate mofetil (MMF) in patients of African ancestry, but MMF is much more expensive than cyclophosphamide and there was low usage of this agent in the studies reviewed by Ameh et al. There was no consistent reporting on the use of adjuvant therapies such as chloroquine. The overall mortality ranged from 7.5% to 34.9%, with a study from Kenya reporting mortality in excess of 80% in patients with proliferative lupus nephritis.[133] The five-year renal survival was 48% to 84%, whereas five-year patient survival was 54% to 94%. Outcomes were better for the studies reported from North Africa. This systematic review has highlighted the severity of SLE and the poor outcomes for lupus nephritis in sub-Saharan Africa.[131]

A recent paper by Tannor et al.[134] examined the utility of repeat renal biopsies in South African patients with lupus nephritis. In 96 patients who had developed a disease flare, repeat biopsies infrequently revealed changes in histologic class and appeared to be of limited value especially when the first biopsy revealed proliferative disease. Even when there were changes in the histologic class, this seldom led to changes in treatment. The authors suggested that repeat biopsies be considered if the reference biopsy is nonproliferative or in the case of nonresponders or partial responders. In this study, there were poor outcomes following the repeat biopsy in patients with disease flares. There was failure to respond to induction therapy in 62.2%; end-stage renal disease developed within one year in 36.3%; and the mortality rate after one year was 23.0%.

SCHISTOSOMIASIS

Schistosoma haematobium or *Schistosoma mansoni* is endemic to many countries of Africa. The disease burden of schistosomiasis in Africa may be equivalent to malaria or HIV/AIDS, and despite the fact that praziquantel is available as a simple annual treatment for less than 50 cents per person, fewer than 5% of the infected population is treated.[135] Terminal hematuria is the typical clinical presentation, usually associated with frequency and dysuria. In a large cross-sectional study in an untreated African population infected with *S. haematobium,* microhematuria was reported in 41% to 100%, and gross hematuria in 0% to 97%.[136]

Although the early renal lesions of schistosomiasis may regress with appropriate antiparasitic treatment, the chronic sequelae are irreversible.[137] Renal failure results from the deposition of ova in the genitourinary system, which causes obstruction, reflux, infection, and stone formation. Immune complexes containing worm antigens may also deposit in the glomeruli, which leads to different patterns of glomerular disease, including mesangioproliferative, exudative, and mesangiocapillary types, focal segmental sclerosing lesions, or amyloidosis.[137,138] Exudative lesions occur in the presence of salmonella coinfection. The development of mesangiocapillary GN and focal segmental sclerosis correlates with the degree of associated hepatic fibrosis; amyloidosis occurs in prolonged infection and correlates with the antigen load.

RENAL STONES

North African countries lie within a geographical Afro-Asian stone-forming belt where renal stone disease is particularly common. In this region, all age groups are affected, with a male predominance. The high prevalence of urolithiasis may be related to high rates of consanguinity, hot climates, and dietary factors such as high intake of animal protein.[139]

There is a paucity of African data on CKD or ESRD resulting from renal stone disease. In Tunisia, 2.7% of patients on chronic dialysis have renal stones as the etiology.[81] In South Africa, only 1.5% of patients on RRT have obstruction and/or reflux as the primary renal disease.[79] Although some of these cases may have been related to renal stones, there were very few patients (24 out of 10,360) where there was specific mention of calculi, uric acid nephropathy, cystinosis, hyperoxaluria, or hypercalcaemia. From the Sudan, Abboud et al. reported that 12% of Sudanese patients with chronic renal failure had renal calculi as the etiology,[140] whereas Elsharif et al. reported obstruction as the etiology of ESRD in 11.6% of patients on chronic hemodialysis.[141] The prevalence of obstructive uropathy in ESRD patients has been reported as 5% in Libya[84] and 11% in Egypt.[142]

SICKLE CELL DISEASE

Africa carries approximately 75% of the global disease burden of sickle cell disease, mainly due to a high prevalence of sickle cell trait in West Africa.[143] It is estimated that 25% of Nigerian adults are carriers. The presence of sickle cell trait protects against the severe complications of *Plasmodium falciparum* malaria, which is endemic in the region.

Renal involvement is common, and more severe in homozygous disease (sickle cell anemia, HbSS) than in the compound heterozygous forms; however, even sickle cell trait is associated with increased risk of CKD.[144] A Nigerian case series by Arogundade et al. reported that 37.4% of their patients with sickle cell disease had CKD as defined by persistent proteinuria, hematuria and/or reduction in GFR.[145]

Hemolysis, vasoocclusion, and ischemia-reperfusion injury are the mechanisms underlying the clinical manifestations of sickle cell disease. The pathophysiology of the renal involvement involves the coexistence of cortical hyperperfusion, medullary hypoperfusion, and an increased renal vasoconstrictive response to systemic and regional stress.[146] Red blood cells sickle in the renal medulla because of its hypoxic, acidotic and hyperosmolar conditions. Clinical manifestations are protean, ranging from hematuria and loss of concentrating ability to acute kidney injury, chronic kidney disease, and renal medullary carcinoma. The most common glomerular lesion seen is focal segmental glomerulosclerosis. Hyperfiltration and impaired concentrating ability may occur in infancy; albuminuria may develop in childhood and progress thereafter; and in older adults the risk of CKD and end-stage renal disease increases. Hematuria and acute kidney disease can occur at any age.[146]

Renal involvement contributes to the diminished life expectancy of patients with sickle cell disease, accounting for 16% to 18% of mortality. Even patients who are able to access RRT still have a mortality rate several-fold higher than patients without sickle cell disease.[146]

HYPERTENSIVE RENAL DISEASE

In registry data from African countries, hypertension has been reported as the etiology of ESRD in 10.1% to 36.6% of patients on RRT (see Table 77.4). Egypt (36.6%) and South Africa (33.7%) have the highest percentages of patients with this primary renal diagnosis, while Morocco (10.1%) and Tunisia (13.6%) are at the lower end of the spectrum.

It is generally accepted that accelerated and malignant forms of hypertension can lead to end-stage renal failure,[147,148] although renal recovery sometimes occurs months after starting RRT, especially when patients are treated with peritoneal dialysis.[149] In a study to determine the basis of ESRD in black South Africans, Gold et al.[150] reviewed the renal histology in 65 black patients on chronic hemodialysis and reported that essential/primary malignant hypertension was the most common cause, occurring in 49%.

Nephrologists also often apply a diagnosis of "hypertensive renal disease" or "hypertensive nephrosclerosis" to nondiabetic patients, especially black patients, who present with chronic kidney disease and only mild-to-moderate hypertension.[78] It has now become clear that many of these patients have glomerular disease, which is related to the presence of renal-risk variants of the APOL1 gene and have secondarily elevated blood pressure.[151] For example, in the African American Study of Kidney Disease and Hypertension (AASK), the APOL1 association was strongest in individuals with more proteinuria and progressive nephropathy. Aggressive hypertension control and the use of angiotensin-converting enzyme inhibitors failed to halt progression, supporting the role of mechanisms other than hypertension.[152] The AASK patients predominantly had focal glomerulosclerosis with interstitial and vascular changes that did not correlate with blood pressure.[153] This new information has prompted calls for the term "hypertensive nephrosclerosis" to be abandoned.[78,154]

The South African Renal Registry has recommended that users submitting data indicate hypertensive nephropathy as the primary renal diagnosis only if there is no reason to suspect that the hypertension may be secondary to preexisting renal disease.[79] They suggest that the following criteria be met: hypertension known to precede renal dysfunction, left ventricular hypertrophy, proteinuria less than 2 g/day, and no evidence of other renal diseases.[155,156]

GENETIC DISORDERS

To date, Africans have only participated minimally in genomics research, and then often only at the level of sample collection. The many reasons for this include a shortage of African scientists with genomic research expertise, lack of research infrastructure, limited computational expertise and resources, and inadequate support by governments.[157] Adedokun et al.[158] performed an analysis of genetics publications from sub-Saharan Africa for the period 2004 to 2013 and found that most of the publications involved populations from South Africa (31.1%), Ghana (10.6%), and Kenya (7.5%). Almost half of all the studies investigated HIV, TB or malaria, while about one-tenth related to NCDs. Less than half of the publications had a first author from an African institution. In most studies without an African first author, local authors mostly had administrative roles and were seldom involved in the conceptualization or design of the studies, data analysis, or in the writing of the manuscripts.[158]

The Human Heredity and Health in Africa (H3Africa) initiative is a large-scale collaboration between the US National Institutes of Health (NIH), the UK-based Wellcome Trust and the African Society of Human Genetics, which aims at addressing some of these issues by focusing strongly on capacity building, as well as specific scientific goals.[157] The project is training African clinical research personnel and genomic investigators, establishing genomic research laboratories in West Africa, and conducting international-level genetic and translational research. Regarding renal disease, the H3Africa Kidney Disease Research Network is currently recruiting patients in Ghana, Nigeria, Ethiopia and Kenya, and analyzing environmental and genetic factors associated with CKD. The recruitment targets are 4000 participants with CKD and 4000 controls, and 50 families with hereditary glomerular disease.[159,160]

Among the genetic disorders relevant to Africa, which are actively researched, is the association of kidney disease with African "risk" variants of the APOL1 gene (see Chapter 43)[160] and genetic factors in sickle cell disease.[161,162]

The Role of Consanguineous Marriage

A consanguineous marriage is defined as a union between two individuals who are related as second cousins or closer. This practice is common throughout Arab countries and contributes to the increase in genetic disorders, especially autosomal recessive disorders, also seen in North Africa (see Chapter 78). For example, the high rates of renal stone disease seen in the region is at least partly due to consanguinity, which results in higher rates of predisposing conditions such as primary hyperoxaluria and cystinosis.[139] Jaouad et al.[163] studied the rates of consanguineous marriages in 176 Moroccan families with autosomal recessive diseases. Although the overall reported prevalence of consanguinity in Morocco was 15%, the rate among the families with autosomal recessive disorders was 59%.

Cystic Kidney Disease

Worldwide, autosomal dominant polycystic kidney disease (ADPKD) is a common cause for CKD and ESRD, but the data from African populations are sparse. Bourquia[164] reported data on ADPKD in 308 Moroccan families. The mean age at diagnosis was 46 years; the most frequent symptom was pain, which led to a diagnosis. Hypertension was present in 11% of the patients, loss of renal function in 17%, and hepatic cysts in 17.8%. In Benin, Laleye et al. described 70 patients with ADPKD.[165] Their mean age was 47.2 years, a positive family history was obtained in 47%, and lumbar pain was the most common presenting symptom. Mutations in the polycystin 1 gene (PKD1) were identified, including several variants not previously described. A cohort of 53 Senegalese patients has also been described,[166] as has a case where a PKD1 mutation was identified.[167] In the Senegalese cohort, there was a high rate of ESRD at the time of diagnosis or soon thereafter, mainly due to delays in making the diagnosis and in referrals to nephrologists. In the Seychelles, ADPKD is seen almost exclusively in people of Caucasian extraction, even though they comprise only 30% of the population. This may be the consequence of a founder effect.[168]

Autosomal recessive polycystic kidney disease (ARPKD) is caused by mutations in the PKHD1 gene and has a worldwide incidence of 1:20,000, corresponding to a heterozygote frequency of 1:70. Among the Afrikaner population of South Africa, the live-birth rate and the carrier rate have been estimated at 1:11,000 and 1:53, respectively.[169] Lambie et al.[170] studied a cohort of patients from 36 Afrikaner families, finding that 27 patients, from 24 families, were homozygous for the p.M627K substitution, providing strong evidence of a founder mutation.

Nephronophthisis (NPHP) is an autosomal recessive cystic kidney disease, which is a leading genetic cause of ESRD in children. In contrast to ADPKD, the kidneys are not enlarged, cysts are present mostly at the corticomedullary junction, and there is prominent tubulointerstitial fibrosis.[171] Soliman et al.[172] studied 20 Egyptian children with NPHP, reporting a mean age at diagnosis of 88 months, and a mean age at the onset of symptoms of 44 months. Patients were categorized as 75% juvenile NPHP, 5% infantile NPHP and 20% Joubert syndrome-related disorders. Homozygous NPHP1 deletions were detected in six patients.

Metabolic Diseases

Primary hyperoxaluria type 1 (PH1) is an autosomal recessive metabolic disease where there is an increased endogenous production of oxalate. Progressive kidney failure results from the excessive urinary oxalate excretion and renal deposition of calcium oxalate. Disease-causing mutations of the alanine glyoxylate aminotransferase (AGXT) gene have been identified in patients from Tunisia[173] and Egypt.[174]

Familial Mediterranean fever (FMF) is an autosomal recessive disease that primarily affects populations surrounding the Mediterranean and it is characterized by recurrent attacks of fever and serosal inflammation. Amyloidosis is a serious complication of the disease and may result in renal failure.

The disease is caused by mutations in the FMF gene (MEFV), which encodes pyrin, a protein involved in regulating inflammatory responses. Mutations may cause gain of function so that pyrin can initiate inflammation in the absence of a toxin or infection. MEFV mutations have been identified in Tunisian patients.[175] Amyloidosis is more common in North African patients, who are homozygous for the M694V mutation; the risk also increases in male patients and in those bearing the polymorphism a/a in the SAA1 gene.[176]

Tubular Disorders

Familial hypomagnesemia with hypercalciuria and nephrocalcinosis caused by a novel claudin16 mutation has been reported in an Egyptian family.[177] In South Africa, new mutations of the SLC12A3 gene, which encodes the sodium-chloride cotransporter on the apical membrane of the distal convoluted tubule, have been reported in a family with Gitelman syndrome, who presented with hypokalaemia and unusual cravings for salt and vinegar.[178] Mutations of the epithelial sodium channel (ENaC) have also been identified in South African patients in association with hypertension[179,180] and preeclampsia.[181] In Tunisia, mutations of the H^+-ATPase pump have been identified in children with primary distal renal tubular acidosis, often associated with sensorineural deafness.[182]

Glomerular Diseases

Mutations of the nephrin gene (NPHS1) have been discovered in North African patients with congenital nephrotic syndrome of the Finnish type.[183] A Moroccan infant with steroid-resistant nephrotic syndrome was found to have a mutation in the ARHGDIA gene, which encodes Rho-GDP dissociation inhibitor 1. This protein modulates signaling through Rho GTPases and thereby regulates cell motility.[184]

CHRONIC KIDNEY DISEASE OF UNCERTAIN ETIOLOGY

CKD of uncertain etiology (CKDu) describes CKD not attributable to traditional risk factors, such as diabetes, hypertension, or HIV infection. CKDu is reported with increasing frequency and has reached epidemic proportions in many countries. The term CKDu is a generic one, and likely includes kidney diseases with many disparate etiologies. It serves to highlight the need for hot-spot identification through adequate surveillance and for collaborative studies to identify specific causes.

The term was first used in El Salvador to describe a disease affecting mainly agricultural communities and has subsequently been reported in many other countries, including Nicaragua, Costa Rica, and Sri Lanka.[185] Factors that have been implicated in the pathogenesis of CKDu include exposure to agricultural toxins, traditional medicines, heavy metals, and heat stress.[185] Interactions between genetic and environmental factors may also be important, as suggested by the finding that a single-nucleotide polymorphism in *SCL13A3* predisposes to CKDu in Sri Lanka.[186]

CKDu hotspots have also been reported from several African countries. For example, in Tunisia, chronic interstitial nephropathy has been linked to the contamination of food with ochratoxin,[187] while in Tanzania, urban residence was a strong risk factor for the presence of CKD.[71] In the El-Minia region of Egypt, ESRD with an unknown etiology in rural farm workers has been associated with environmental factors, such as drinking unsafe water, exposure to pesticides, and using herbal medications.[188]

SCREENING AND PREVENTION OF CHRONIC KIDNEY DISEASE

CKD can be prevented by effective control of known etiologies and risk factors and by implementing adequate renoprotective measures. The early detection of CKD by efficient screening programs is important, especially in African countries where renal replacement therapy is not easily accessible.[189]

Various challenges need to be overcome to include CKD screening and prevention in daily clinical practice. The access to diagnostic tools is sometimes a challenge for practitioners and patients. Ensuring the accuracy of the diagnosis and confirming the chronicity of the renal abnormalities is a necessity, since the classification of a person as having CKD has major consequences.[76] In the Maremar study,[190] the investigators demonstrated that repeating eGFR after 3 months avoided a false positive diagnosis of CKD in 32% of cases. The same reasoning can be applied for albuminuria and proteinuria, the other common screening tests for CKD. Repeating dipstick testing after 1 to 2 weeks allowed positive results to be identified as false positives in 67.5% of subjects with mild proteinuria (dipstick +) and in 28.7% of those with overt proteinuria (dipstick ++ to +++).[190] Another recent African epidemiological study, which determined the prevalence of CKD in sugarcane workers in Cameroon, also included the element of chronicity in their methods, and reported a relatively low CKD rate of 3.4%.[191]

Screening mostly detects individuals with earlier stages of CKD, and therefore the CKD-EPI study equation is recommended, as it is more accurate than the MDRD equation at higher GFR values. As discussed in the Chronic Kidney Disease section previously, the ethnicity correction factor should not be used.[61–63]

Another challenge is the feasibility and the cost-effectiveness of CKD screening and prevention programs in African countries. Many epidemiological studies have shown that diseases such as diabetes, hypertension, and HIV infection are the major causes of CKD in many countries.[67,68,192] It seems reasonable to couple simple CKD screening and preventive measures with already-implemented programs for diabetes, hypertension, and HIV infection.[193] For example, the WHO STEPS method is a simple, standardized method for collecting data on NCDs and their risk factors, and could easily be adapted to include screening for CKD.[194]

Education of patients and healthcare workers regarding hypertension, diabetes, obesity, and healthy lifestyles (e.g., prudent diet, cessation of smoking, exercise) is essential for primary prevention of CKD. The programs of organizations such as the ISN and initiatives such as World Kidney Day have heightened awareness of renal disease among both medical professionals and the public in developing countries. Efforts should be made to optimize therapy for hypertension, diabetes mellitus, and renal disease, applying best practice clinical guidelines for the management of these conditions. Unfortunately, the shortage of healthcare workers and the lack of availability of pharmaceutical agents to retard progression of CKD are impediments in many African countries.[195]

END-STAGE RENAL DISEASE AND RENAL REPLACEMENT THERAPY

Anand et al.[196] have predicted an annual ESRD incidence of 239 per million population (PMP) in people with diabetes and hypertension who live in sub-Saharan Africa. In North African countries, the annual incidence has been estimated at 150 PMP.[197,198] The wide variations in the reported prevalence of treated ESRD seen in developing countries are mainly a reflection of differing access to dialysis and transplantation rather than differences in the burden of disease,[199] and this provision of RRT is strongly associated with gross national income per capita.[200] Liyanage et al.[201] have estimated that at least 432,000 people in Africa require RRT but are not receiving it. In most countries where RRT is available, patients carry the cost, and few are able to afford dialysis beyond the first 3 months.[202]

The lack of renal registries means that there are few reliable, current statistics on RRT in Africa.[203] Estimates are often based on old registry reports and unpublished data,[204,205] but suggest that the provision of RRT services has been a low priority for the governments of most African countries. A few renal registries, mainly from North Africa, have been established in African countries but these projects have mostly not been sustainable, and reports have not been published regularly. The earliest reports were from Egypt and Tunisia in 1975, followed by South Africa in 1977, and thereafter by Libya, Algeria, and Morocco.[203]

Table 77.6 summarizes the African publications since 1996 that have most recently reported country wide data on RRT.[203] Data from North African countries also have been published in the European Renal Association–European Dialysis and Transplant Association (ERA-EDTA) Registry Annual Reports over a number of years.[203]

South Africa is the only country on the continent that has regularly reported country wide data on RRT in recent years.[79,206–208] In December 2015, the number of patients who were treated with chronic dialysis or transplantation in South Africa stood at 10,360, a prevalence of 189 PMP. The treatment modality was hemodialysis in 72.7%, peritoneal dialysis in 13.9%, and transplantation in 13.4%. The prevalence, which was 167 PMP in 2013 and 178 PMP in 2014, has been increasing due to increasing numbers of patients accessing hemodialysis in the private healthcare sector. In the public sector, which serves 84% of the South African population, the prevalence of RRT was 71.9 PMP, as compared with 799.3 PMP in the private sector. Disparities between different provinces were also identified as were disparities related to ethnicity, with Blacks being the most underserved group.[79]

The African Association of Nephrology (AFRAN) and the African Pediatric Nephrology Association (AFPNA) are now addressing this serious knowledge gap and have established the African Renal Registry, using the expanded platform of the South African Renal Registry.[209] Ghana, Zambia, Burundi, and Kenya have formally joined the African Renal Registry during this pilot phase.

Recently, data on RRT from Egypt and Morocco have been published in the International Comparisons chapter of the United States Renal Data System Annual Data Report.[210] In Egypt, the prevalence of RRT in 2015 was 624 PMP. The incidence of new patients starting RRT was 56 PMP, with diabetes the cause of the renal disease in 15%. The 2015 transplant rate was 15 PMP. In Morocco, the prevalence of RRT in 2015 was 541 PMP and the incidence 144 PMP. Diabetes was reported as the cause of the renal disease in 44% of the patients. The transplant rate was 1 PMP.

There have been few economic evaluation studies of RRT in African countries. At Muhimbili National Hospital in Tanzania, Mushi et al.[211] estimated the costs of hemodialysis for 34 patients in 2014 to be US$176 per treatment, or US$27,440 per annum. In Cameroon, Halle et al.[212] conducted a cost analysis of hemodialysis care in 154 patients at three public-sector facilities. The annual cost of hemodialysis per patient was US$13,581, with out-of-pocket payments of US$4,114, accounting for 30% of the total cost. The authors concluded that the cost of maintaining patients on hemodialysis was very high considering the national gross domestic

Table 77.6 Most Recent Publications Reporting Countrywide Data on Renal Replacement Therapy From African Countries

Author and Year	Country/Region	Last Data	Key Data Reported
USRDS Report, 2017[210]	Egypt	2015	Incidence 56 PMP, diabetes the cause in 15%; prevalence 624 PMP. Dialysis modality: HD 100%. Transplant rate 15 PMP, all living donors.
Alashek et al., 2012[84]	Libya	2010	Dialysis-treated ESRD. Incidence 282 PMP, diabetes the cause in 28.4%; prevalence 624 PMP, diabetes the cause in 26.5%, unknown cause in 7.3%.
USRDS Report, 2017[210]	Morocco	2015	Incidence 144 PMP, diabetes the cause in 44%; prevalence 541 PMP. Transplant rate 1 PMP.
Albitar et al., 1998[85]	Reunion Island	1996	Incidence 188 PMP, prevalence 1155 PMP, cause was diabetes in 33.6%. Modality: HD 82.7%, PD 3.0%, TX 14.3%. Annual mortality 8.1%.
Davids et al., 2017[79]	South Africa	2015	Prevalence 189 PMP, cause unknown in 34.1%, hypertension in 33.7%, diabetes in 14.4%. Modality: HD 72.7%, PD 13.9%, TX 13.4%. Transplant rate 4.6 PMP.
Elamin et al., 2010[80]	Sudan	2009	Prevalence 106 PMP, no cause identified in 42.6%, diabetes in 10.4%. Modality: HD 68.9%, PD 2.9%, TX 28.2%.
Ministry of Health, 2010[81]	Tunisia	2010	Incidence 133 PMP; prevalence 806 PMP, diabetes the cause in 21.0%. Transplants performed during 2010 = 132, living donors in 77.3%.

ESRD, End stage renal disease; *HD,* hemodialysis; *PD,* peritoneal dialysis; *PMP,* per million population; *TX,* transplant.

product per capita in Cameroon, and that the out-of-pocket payments were unsustainable for most patients.

The outcomes of patients requiring chronic dialysis in sub-Saharan Africa have been reviewed by Ashuntantang et al.[213] Summarizing 68 studies, they reported a treatment discontinuation rate of 59% in adults and 49% in children. This attrition usually occurred within the first two weeks of starting dialysis and was mainly the result of an inability to pay for the treatment. When studies from South Africa and Sudan were excluded, the proportions of patients who discontinued dialysis were 78% and 86% among adults and children, respectively. When patients were able to pay for long-term dialysis, outcomes were improved, with over 75% remaining on dialysis for over one year.

Several African studies have examined the quality of life (QOL) of patients on RRT. These have mainly used standardized questionnaires such as the Kidney Disease Quality of Life–Short Form (KDQOL-SF) and the KDQOL-36 questionnaire. Okaka et al.[214,215] found that patients on CAPD in Johannesburg, South Africa, had lower QOL scores than healthy controls, with better scores in patients under the age of 30 years, those with a duration on PD of less than 4 years and those with higher incomes. In Cape Town, Tannor et al.[216] used mixed methods to investigate the QOL of patients on chronic hemodialysis and peritoneal dialysis and found that PD patients experienced a heavier symptom burden and greater limitations related to their dialysis modality, especially with regards to social functioning. The fear of developing peritonitis caused many patients to sacrifice social interactions to be able do their PD exchanges at home and at the specified times. In Malawi, Masina et al.[217] found the QOL scores of their hemodialysis patients to be generally low, and particularly low in the case of physical health scores. Similar findings were reported from Kenyatta National Hospital in Nairobi, Kenya, by Kamau et al.[218] In Morocco, El Filali et al.[219] reported high rates of major depressive and anxiety disorders (34% and 25.2%, respectively) in chronic hemodialysis patients. Major depressive episodes were associated with living alone, the presence of pain, and anxiety disorders. Bouidida et al.[220] have developed a Moroccan version of the KDQOL-SF and demonstrated its validity and reliability as a measure of QOL in Moroccan hemodialysis patients. In Egypt, Abdelfatah et al.[221] studied dialysis and transplant recipients, finding that younger patients had better QOL scores and that there was no difference in overall QOL with different modalities of RRT.

CONCLUDING REMARKS

Africa faces many health challenges as its rapidly-growing populations struggle with the multiple burdens of infections, noncommunicable diseases, pregnancy-related diseases and injuries. These factors are the driving force behind an epidemic of acute and chronic kidney disease, which often affects young, economically active individuals and has serious economic consequences for their families and communities. In most African countries, financial and human resources are inadequate to meet this challenge.

There are, however, several successful initiatives which provide hope for the future. Regarding human resources, the ISN and ISPD Fellowship programs are making a big contribution to the training of nephrology healthcare workers in Africa. Events such as World Kidney Day are creating increased awareness of kidney disease among the public and other health professionals, and the outcomes for patients with AKI are improving as a result of the Saving Young Lives and 0by25 projects. Finally, ambitious new research projects such as the H3Africa project and the African Renal Registry have the potential to improve the outcomes of people with kidney disease and provide valuable training platforms for African researchers.

Complete reference list available at ExpertConsult.com.

KEY REFERENCES

2. *World Population Prospects: The 2017 revision, key findings and advance tables: United Nations, Department of Economic and Social Affairs, Population Division*; 2017.
6. *Noncommunicable diseases among young people in Africa*. Washington, DC: Population Reference Bureau; 2015.
8. Marquez PV, Farrington JL. *The challenge of non-communicable diseases and road traffic injuries in Sub-Saharan Africa: an overview*. Washington, DC: The World Bank; 2013.
13. Sharif MU, Elsayed ME, Stack AG. The global nephrology workforce: emerging threats and potential solutions. *Clin Kidney J*. 2016;9:11–22.
16. Harris DC, Dupuis S, Couser WG, et al. Training nephrologists from developing countries: does it have a positive impact? *Kidney Int Suppl (2011)*. 2012;2:275–278.
18. Mehta RL, Burdmann EA, Cerda J, et al. Recognition and management of acute kidney injury in the International Society of Nephrology 0by25 Global Snapshot: a multinational cross-sectional study. *Lancet*. 2016;387:2017–2025.
20. Olowu WA, Niang A, Osafo C, et al. Outcomes of acute kidney injury in children and adults in sub-Saharan Africa: a systematic review. *Lancet Glob Health*. 2016;4:e242–e250.
24. Gerntholtz TE, Goetsch SJ, Katz I. HIV-related nephropathy: a South African perspective. *Kidney Int*. 2006;69:1885–1891.
26. Wearne N, Swanepoel CR, Boulle A, et al. The spectrum of renal histologies seen in HIV with outcomes, prognostic indicators and clinical correlations. *Nephrol Dial Transplant*. 2012;27:4109–4118.
33. Luyckx VA, Naicker S. Acute kidney injury associated with the use of traditional medicines. *Nat Rev Nephrol*. 2008;4:664.
34. Stanifer JW, Kilonzo K, Wang D, et al. Traditional medicines and kidney disease in low-and middle-income countries: opportunities and challenges. *Semin Nephrol*. 2017;37:245–259.
40. Hamdouk MI, Abdelraheem MB, Taha AA, et al. Paraphenylene diamine hair dye poisoning. In: De Broe ME, Porter GA, Bennett WM, et al, eds. *Clinical Nephrotoxins*. 3rd ed. Boston: Springer; 2008:871–879.
42. Chippaux JP. Snake-bites: appraisal of the global situation. *Bull World Health Organ*. 1998;76:515–524.
45. Arrayhani M, El Youbi R, Sqalli T. Pregnancy-related acute kidney injury: experience of the nephrology unit at the University Hospital of Fez, Morocco. *ISRN Nephrol*. 2013;2013:109034.
51. Abdou N, Antwi S, Koffi LA, et al. Peritoneal dialysis to treat patients with acute kidney injury-the Saving Young Lives experience in West Africa: Proceedings of the Saving Young Lives session at the First International Conference of Dialysis in West Africa, Dakar, Senegal, December 2015. *Perit Dial Int*. 2017;37:155–158.
61. van Deventer HE, George JA, Paiker JE, et al. Estimating glomerular filtration rate in black South Africans by use of the Modification of Diet in Renal Disease and Cockcroft-Gault equations. *Clin Chem*. 2008;54:1197–1202.
62. Madala ND, Nkwanyana N, Dubula T, et al. Predictive performance of eGFR equations in South Africans of African and Indian ancestry compared with ^{99m}Tc-DTPA imaging. *Int Urol Nephrol*. 2012;44:847–855.
63. Sagou Yayo É, Aye M, Konan J-L, et al. Inadequacy of the African-American ethnic factor to estimate glomerular filtration rate in an African general population: Results from Côte d'Ivoire. *Nephrol Ther*. 2016;12:454–459.
67. Benghanem Gharbi M, Elseviers M, Zamd M, et al. Chronic kidney disease, hypertension, diabetes, and obesity in the adult population

of Morocco: how to avoid “over”- and “under”-diagnosis of CKD. *Kidney Int.* 2016;89:1363–1371.

68. Stanifer JW, Jing B, Tolan S, et al. The epidemiology of chronic kidney disease in sub-Saharan Africa: a systematic review and meta-analysis. *Lancet Glob Health.* 2014;2:e174–e181.
77. Kayange NM, Smart LR, Tallman JE, et al. Kidney disease among children in sub-Saharan Africa: systematic review. *Pediatr Res.* 2014;77:272–281.
79. Davids MR, Marais N, Jacobs JC. South African Renal Registry annual report 2015. *Afr J Nephrol.* 2017;20:201–2013.
80. Elamin S, Obeid W, Abu-Aisha H. Renal replacement therapy in Sudan, 2009. *Arab J Nephrol Transplant.* 2010;3:31–36.
84. Alashek WA, McIntyre CW, Taal MW. Epidemiology and aetiology of dialysis-treated end-stage kidney disease in Libya. *BMC Nephrol.* 2012;13:1.
85. Albitar S, Bourgeon B, Genin R, et al. Epidemiology of end-stage renal failure in Reunion Island (results from the registry of the Indian Ocean Society of Nephrology). *Nephrol Dial Transplant.* 1998;13: 1143–1145.
86. Counil É, Cherni N, Kharrat M, et al. Trends of incident dialysis patients in Tunisia between 1992 and 2001. *Am J Kidney Dis.* 2008;51: 463–470.
89. Noubiap J, Naidoo J, Kengne AP. Diabetic nephropathy in Africa: a systematic review. *World J Diabetes.* 2015;6:759–773.
90. Okpechi IG, Ameh OI, Bello AK, et al. Epidemiology of histologically proven glomerulonephritis in Africa: a systematic review and meta-analysis. *PLoS ONE.* 2016;11:e0152203.
91. Ehrich JH, Eke FU. Malaria-induced renal damage: facts and myths. *Pediatr Nephrol.* 2007;22:626–637.
100. Rosenberg AZ, Naicker S, Winkler CA, et al. HIV-associated nephropathies: epidemiology, pathology, mechanisms and treatment. *Nat Rev Nephrol.* 2015;11:150–160.
102. Kasembeli AN, Duarte R, Ramsay M, et al. APOL1 risk variants are strongly associated with HIV-associated nephropathy in black South Africans. *J Am Soc Nephrol.* 2015.
105. Koech M, Owiti M, Owino-Ong’or W, et al. Absence of HIV-associated nephropathy among antiretroviral naïve adults with persistent albuminuria in Western Kenya. *Kidney Int Rep.* 2017;2: 159–164.
111. Fabian J, Naicker S, Goetsch S, et al. The clinical and histological response of HIV-associated kidney disease to antiretroviral therapy in South Africans. *Nephrol Dial Transplant.* 2013;28:1543–1554.
123. Bates WD *Hepatitis-B-associated glomerular disease: a clinicopathological study of hepatitis B virus associated membranous glomerulonephritis in Namibian and South African children 1974-2005 and a comparison with hepatitis B associated membranous glomerulonephritis as well as idiopathic membranous glomerulonephritis in adults.* Cape Town: Stellenbosch University; 2011.
131. Ameh O, Kengne A, Jayne D, et al. Standard of treatment and outcomes of adults with lupus nephritis in Africa: a systematic review. *Lupus.* 2016;25:1269–1277.
137. Barsoum RS. Schistosomiasis and the kidney. *Semin Nephrol.* 2003; 23:34–41.
146. Nath KA, Hebbel RP. Sickle cell disease: renal manifestations and mechanisms. *Nat Rev Nephrol.* 2015;11:161–171.
152. Freedman BI, Langefeld CD. The new era of APOL1-associated glomerulosclerosis. *Nephrol Dial Transplant.* 2012;27:1288–1291.
158. Adedokun BO, Olopade CO, Olopade OI. Building local capacity for genomics research in Africa: recommendations from analysis of publications in Sub-Saharan Africa from 2004 to 2013. *Glob Health Action.* 2016;9:31026.
160. Osafo C, Raji YR, Olanrewaju T, et al. Genomic approaches to the burden of kidney disease in Sub-Saharan Africa: the Human Heredity and Health in Africa (H3Africa) Kidney Disease Research Network. *Kidney Int.* 2016;90:2–5.
185. Lunyera J, Mohottige D, Von Isenburg M, et al. CKD of uncertain etiology: a systematic review. *Clin J Am Soc Nephrol.* 2015;11:1–7.
196. Anand S, Bitton A, Gaziano T. The gap between estimated incidence of end-stage renal disease and use of therapy. *PLoS ONE.* 2013;8:e72860.
201. Liyanage T, Ninomiya T, Jha V, et al. Worldwide access to treatment for end-stage kidney disease: a systematic review. *Lancet.* 2015;385: 1975–1982.
202. Arogundade F, Sanusi A, Hassan M, et al. The pattern, clinical characteristics and outcome of ESRD in Ile-Ife, Nigeria: Is there a change in trend? *Afr Health Sci.* 2011;11:594–601.
203. Davids MR, Caskey FJ, Young T, et al. Strengthening renal registries and ESRD research in Africa. *Semin Nephrol.* 2017;37:211–223.
209. Davids MR, Eastwood JB, Selwood NH, et al. A renal registry for Africa: first steps. *Clin Kidney J.* 2016;9:162–167.
211. Mushi L, Krohn M, Flessa S. Cost of dialysis in Tanzania: evidence from the provider’s perspective. *Health Econ Rev.* 2015;5:28.
212. Halle MP, Jimkap NN, Kaze FF, et al. Cost of care for patients on maintenance haemodialysis in public facilities in Cameroon. *Afr J Nephrol.* 2017;20:230–237.
213. Ashuntantang G, Osafo C, Olowu WA, et al. Outcomes in adults and children with end-stage kidney disease requiring dialysis in sub-Saharan Africa: a systematic review. *Lancet Glob Health.* 2017;5: e408–e417.
216. Tannor EK, Archer E, Kapembwa K, et al. Quality of life in patients on chronic dialysis in South Africa: a comparative mixed methods study. *BMC Nephrol.* 2017;18:4.

REFERENCES

1. *2017 World Population Data Sheet*; 2017.
2. *World Population Prospects: The 2017 revision, key findings and advance tables: United Nations, Department of Economic and Social Affairs, Population Division*; 2017.
3. *The World's youth 2013 data sheet*. Washington, DC: Population Reference Bureau; 2013.
4. *Africa's Pulse, No. 16*. Washington, DC: World Bank; 2017.
5. World Health Organization. *World health statistics 2016: monitoring health for the SDGs: World Health Organization*; 2016.
6. *Noncommunicable diseases among young people in Africa*. Washington, DC: Population Reference Bureau; 2015.
7. Mayosi BM, Flisher AJ, Lalloo UG, et al. The burden of non-communicable diseases in South Africa. *Lancet*. 2009;374:934–947.
8. Marquez PV, Farrington JL. *The challenge of non-communicable diseases and road traffic injuries in Sub-Saharan Africa: an overview*. Washington, DC: The World Bank; 2013.
9. *Road safety in the WHO African Region. The Facts 2013*. Geneva, Switzerland: World Health Organization; 2013.
10. *Global health sector response to HIV, 2000-2015*. Geneva, Switzerland: World Health Organization; 2015.
11. *Global health observatory (GHO) data. Density of physicians (total number per 1000 population, latest available year)*. World Health Organization; 2018. At: http://www.who.int/gho/health_workforce/physicians_density_text/en/. Accessed February 5, 2018.
12. Benghanem Gharbi M *Chronic dialysis programs in African countries: Barriers and successes*. Cape Town, South Africa: World Congress of Nephrology; 2015.
13. Sharif MU, Elsayed ME, Stack AG. The global nephrology workforce: emerging threats and potential solutions. *Clin Kidney J*. 2016;9:11–22.
14. Mandayam S, Winkelmayer WC. Worldwide preparedness for kidney health care. *JAMA*. 2017;317:1838–1839.
15. Eastwood JB, Conroy RE, Naicker S, et al. Loss of health professionals from sub-Saharan Africa: the pivotal role of the UK. *Lancet*. 2005;365:1893–1900.
16. Harris DC, Dupuis S, Couser WG, et al. Training nephrologists from developing countries: does it have a positive impact? *Kidney Int Suppl (2011)*. 2012;2:275–278.
17. Jha V, Arici M, Collins AJ, et al. Understanding kidney care needs and implementation strategies in low- and middle-income countries: conclusions from a "Kidney Disease: Improving Global Outcomes" (KDIGO) Controversies Conference. *Kidney Int*. 2016;90:1164–1174.
18. Mehta RL, Burdmann EA, Cerda J, et al. Recognition and management of acute kidney injury in the International Society of Nephrology 0by25 Global Snapshot: a multinational cross-sectional study. *Lancet*. 2016;387:2017–2025.
19. Mehta RL, Macedo E, Zhang J, *ISN 0by25 Initiative Steering Committee. International Society of Nephrology 0by25 Global Snapshot*. Regional report for Africa; 2015.
20. Olowu WA, Niang A, Osafo C, et al. Outcomes of acute kidney injury in children and adults in sub-Saharan Africa: a systematic review. *Lancet Glob Health*. 2016;4:e242–e250.
21. Bourial M. *Aspects epidemiologiques de l'insuffisance renale aigue au Maroc. Premiere enquete nationale*. Casablanca, Morocco: Université Hassan II de Casablanca; 2014.
22. Wyatt CM, Arons RR, Klotman PE, et al. Acute renal failure in hospitalized patients with HIV: risk factors and impact on in-hospital mortality. *AIDS*. 2006;20:561–565.
23. Boswell MT, Rossouw TM. Approach to acute kidney injury in HIV-infected patients in South Africa. *South Afr J HIV Med*. 2017;18:a714.
24. Gerntholtz TE, Goetsch SJ, Katz I. HIV-related nephropathy: a South African perspective. *Kidney Int*. 2006;69:1885–1891.
25. Vermeulen A, Naicker S *A review of patterns of renal disease at Chris Hani Baragwanath Academic Hospital from 1982 to 2011*. Johannesburg, South Africa: University of the Witwatersrand; 2014.
26. Wearne N, Swanepoel CR, Boulle A, et al. The spectrum of renal histologies seen in HIV with outcomes, prognostic indicators and clinical correlations. *Nephrol Dial Transplant*. 2012;27:4109–4118.
27. Vachiat AI, Musenge E, Wadee S, et al. Renal failure in HIV-positive patients-a South African experience. *Clin Kidney J*. 2013;6:584–589.
28. Arendse C, Okpechi I, Swanepoel C. Acute dialysis in HIV-positive patients in Cape Town, South Africa. *Nephrology*. 2011;16:39–44.
29. Camara SN, Chothia M-Y. Outcomes of critically ill adult patients with continuous renal replacement therapy-requiring acute kidney injury in the Free State province of South Africa: the impact of HIV. *Afr J Nephrol*. 2017;20:39–44.
30. Organization WH. *World Malaria Report 2017*. Geneva, Switzerland: World Health Organization; 2017.
31. Jha V, Parameswaran S. Community-acquired acute kidney injury in tropical countries. *Nat Rev Nephrol*. 2013;9:278–290.
32. Mishra SK, Das BS. Malaria and acute kidney injury. *Semin Nephrol*. 2008;28:395–408.
33. Luyckx VA, Naicker S. Acute kidney injury associated with the use of traditional medicines. *Nat Rev Nephrol*. 2008;4:664.
34. Stanifer JW, Kilonzo K, Wang D, et al. Traditional medicines and kidney disease in low-and middle-income countries: opportunities and challenges. *Semin Nephrol*. 2017;37:245–259.
35. Luyckx VA. Nephrotoxicity of alternative medicine practice. *Adv Chronic Kidney Dis*. 2012;19:129–141.
36. Pozdzik A, Salmon I, Debelle F, et al. Aristolochic acid induces proximal tubule apoptosis and epithelial to mesenchymal transformation. *Kidney Int*. 2008;73:595–607.
37. Stewart MJ, Steenkamp V. The biochemistry and toxicity of atractyloside: a review. *Ther Drug Monit*. 2000;22:641–649.
38. Fang H-C, Lee P-T, Lu P-J, et al. Mechanisms of star fruit-induced acute renal failure. *Food Chem Toxicol*. 2008;46:1744–1752.
39. Suliman SM, Homeida M, Aboud OI. Paraphenylenediamine induced acute tubular necrosis following hair dye ingestion. *Hum Toxicol*. 1983;2:633–635.
40. Hamdouk MI, Abdelraheem MB, Taha AA, et al. Paraphenylene diamine hair dye poisoning. In: De Broe ME, Porter GA, Bennett WM, et al, eds. *Clinical Nephrotoxins*. 3rd ed. Boston, MA: Springer; 2008:871–879.
41. Warrell DA. Snake bite. *Lancet*. 2010;375:77–88.
42. Chippaux JP. Snake-bites: appraisal of the global situation. *Bull World Health Organ*. 1998;76:515–524.
43. Chippaux J-P, Goyffon M. Epidemiology of scorpionism: a global appraisal. *Acta Trop*. 2008;107:71–79.
44. Isbister GK, Bawaskar HS. Scorpion envenomation. *N Engl J Med*. 2014;371:457–463.
45. Arrayhani M, El Youbi R, Sqalli T. Pregnancy-related acute kidney injury: experience of the nephrology unit at the University Hospital of Fez, Morocco. *ISRN Nephrol*. 2013;2013:109034.
46. Okafor UV, Efetie RE. Critical care management of eclamptics: challenges in an African setting. *Trop Doct*. 2008;38:11–13.
47. Mjahed K, Alaoui SY, Barrou L. Acute renal failure during eclampsia: incidence risks factors and outcome in intensive care unit. *Ren Fail*. 2004;26:215–221.
48. Olowu WA. Acute kidney injury in children in Nigeria. *Clin Nephrol*. 2015;83:70–74.
49. Mehta RL, Cerda J, Burdmann EA, et al. International Society of Nephrology's 0by25 initiative for acute kidney injury (zero preventable deaths by 2025): a human rights case for nephrology. *Lancet*. 2015;385:2616–2643.
50. Smoyer WE, Finkelstein FO, McCulloch MI, et al. "Saving Young Lives" with acute kidney injury: the challenge of acute dialysis in low-resource settings. *Kidney Int*. 2016;89:254–256.
51. Abdou N, Antwi S, Koffi LA, et al. Peritoneal dialysis to treat patients with acute kidney injury-the Saving Young Lives experience in West Africa: Proceedings of the Saving Young Lives session at the First International Conference of Dialysis in West Africa, Dakar, Senegal, December 2015. *Perit Dial Int*. 2017;37:155–158.
52. Ponce D, Balbi A, Cullis B, et al. Evidence, guidelines, and controversies. *Semin Nephrol*. 2017;37:103–112.
53. Cullis B, Abdelraheem M, Abrahams G, et al. Peritoneal dialysis for acute kidney injury. *Perit Dial Int*. 2014;34:494–517.
54. Naicker S, Aboud O, Gharbi MB. Epidemiology of acute kidney injury in Africa. *Semin Nephrol*. 2008;28:348–353.
55. Seedat YK, Nathoo BC. Acute renal failure in blacks and Indians in South Africa-comparison after 10 years. *Nephron*. 1993;64:198–201.
56. Hill NR, Fatoba ST, Oke JL, et al. Global prevalence of chronic kidney disease–a systematic review and meta-analysis. *PLoS ONE*. 2016;11:e0158765.
57. Garcia-Garcia G, Jha V, Tao Li PK, et al. Chronic kidney disease (CKD) in disadvantaged populations. *Clin Kidney J*. 2015;8:3–6.
58. Cass A, Cunningham J, Snelling P, et al. Exploring the pathways leading from disadvantage to end-stage renal disease for indigenous Australians. *Soc Sci Med*. 2004;58:767–785.
59. Eknoyan G, Lameire N. KDIGO 2012 clinical practice guideline for the evaluation and management of chronic kidney disease. *Kidney Int Suppl*. 2013;3:1–150.

60. Coresh J, Astor BC, McQuillan G, et al. Calibration and random variation of the serum creatinine assay as critical elements of using equations to estimate glomerular filtration rate. *Am J Kidney Dis*. 2002;39:920–929.
61. van Deventer HE, George JA, Paiker JE, et al. Estimating glomerular filtration rate in black South Africans by use of the Modification of Diet in Renal Disease and Cockcroft-Gault equations. *Clin Chem*. 2008;54:1197–1202.
62. Madala ND, Nkwanyana N, Dubula T, et al. Predictive performance of eGFR equations in South Africans of African and Indian ancestry compared with ^{99m}Tc-DTPA imaging. *Int Urol Nephrol*. 2012;44: 847–855.
63. Sagou Yayo É, Aye M, Konan J-L, et al. Inadequacy of the African-American ethnic factor to estimate glomerular filtration rate in an African general population: Results from Côte d'Ivoire. *Nephrol Ther*. 2016;12:454–459.
64. Levey AS, Stevens LA, Schmid CH, et al. A new equation to estimate glomerular filtration rate. *Ann Intern Med*. 2009;150:604–612.
65. Pottel H, Hoste L, Delanaye P. Abnormal glomerular filtration rate in children, adolescents and young adults starts below 75 mL/ min/1.73 m2. *Pediatr Nephrol*. 2015;30:821–828.
66. Glassock RJ. Con: Thresholds to define chronic kidney disease should not be age dependent. *Nephrol Dial Transplant*. 2014;29: 774–779.
67. Benghanem Gharbi M, Elseviers M, Zamd M, et al. Chronic kidney disease, hypertension, diabetes, and obesity in the adult population of Morocco: how to avoid "over"- and "under"-diagnosis of CKD. *Kidney Int*. 2016;89:1363–1371.
68. Stanifer JW, Jing B, Tolan S, et al. The epidemiology of chronic kidney disease in sub-Saharan Africa: a systematic review and meta-analysis. *Lancet Glob Health*. 2014;2:e174–e181.
69. Matsha TE, Yako YY, Rensburg MA, et al. Chronic kidney diseases in mixed ancestry South African populations: prevalence, determinants and concordance between kidney function estimators. *BMC Nephrol*. 2013;14:75.
70. Adeniyi AB, Laurence CE, Volmink JA, et al. Prevalence of chronic kidney disease and association with cardiovascular risk factors among teachers in Cape Town, South Africa. *Clin Kidney J*. 2017.
71. Stanifer JW, Maro V, Egger J, et al. The epidemiology of chronic kidney disease in Northern Tanzania: a population-based survey. *PLoS ONE*. 2015;10:e0124506.
72. Seck SM, Doupa D, Guéye L, et al. Epidemiology of chronic kidney disease in northern region of Senegal: a community based study in. *Pan Afr Med J*. 2012;2014:18.
73. Kaze FF, Halle M-P, Mopa HT, et al. Prevalence and risk factors of chronic kidney disease in urban adult Cameroonians according to three common estimators of the glomerular filtration rate: a cross-sectional study. *BMC Nephrol*. 2015;16:96.
74. Kalyesubula R, Nankabirwa JI, Ssinabulya I, et al. Kidney disease in Uganda: a community based study. *BMC Nephrol*. 2017;18:116.
75. Wokoma FS, Emem-Chioma PC, Oko-Jaja RI. Systematic analysis of community studies of risk factors and prevalence of chronic kidney disease in Nigeria (2006-2014). *Tropical Journal of Nephrology*. 2017;11:7–16.
76. Delanaye P, Glassock RJ, De Broe ME. Epidemiology of chronic kidney disease: think (at least) twice! *Clin Kidney J*. 2017.
77. Kayange NM, Smart LR, Tallman JE, et al. Kidney disease among children in sub-Saharan Africa: systematic review. *Pediatr Res*. 2014;77:272–281.
78. Freedman BI, Cohen AH. Hypertension-attributed nephropathy: what's in a name? *Nat Rev Nephrol*. 2016;12:27–36.
79. Davids MR, Marais N, Jacobs JC. South African Renal Registry annual report 2015. *Afr J Nephrol*. 2017;20:201–213.
80. Elamin S, Obeid W, Abu-Aisha H. Renal replacement therapy in Sudan, 2009. *Arab J Nephrol Transplant*. 2010;3:31–36.
81. *Rapport D'Activite Des Centres et Unites D'Hemodialyse, Annee 2010: Ministere de la Sante Publique, Tunisia; 2011 August 2011.*
82. *Insuffisance rénale chronique terminale: Moroccan Society of Nephrology and Moroccan Ministry of Health*; 2013.
83. *Egyptian Renal Registry 9th Annual Report*. Egyptian Society of Nephrology and Transplantation; 2008. At: http://www.esntonline.com/content/downloads/registry/2008.pdf. Accessed August 22, 2016.
84. Alashek WA, McIntyre CW, Taal MW. Epidemiology and aetiology of dialysis-treated end-stage kidney disease in Libya. *BMC Nephrol*. 2012;13:1.
85. Albitar S, Bourgeon B, Genin R, et al. Epidemiology of end-stage renal failure in Reunion Island (results from the registry of the Indian Ocean Society of Nephrology). *Nephrol Dial Transplant*. 1998;13:1143–1145.
86. Counil É, Cherni N, Kharrat M, et al. Trends of incident dialysis patients in Tunisia between 1992 and 2001. *Am J Kidney Dis*. 2008; 51:463–470.
87. Ogurtsova K, da Rocha Fernandes J, Huang Y, et al. IDF Diabetes Atlas: Global estimates for the prevalence of diabetes for 2015 and 2040. *Diabetes Res Clin Pract*. 2017;128:40–50.
88. Gregg EW, Li Y, Wang J, et al. Changes in diabetes-related complications in the United States, 1990-2010. *N Engl J Med*. 2014;370:1514–1523.
89. Noubiap J, Naidoo J, Kengne AP. Diabetic nephropathy in Africa: a systematic review. *World J Diabetes*. 2015;6:759–773.
90. Okpechi IG, Ameh OI, Bello AK, et al. Epidemiology of histologically proven glomerulonephritis in Africa: a systematic review and meta-analysis. *PLoS ONE*. 2016;11:e0152203.
91. Ehrich JH, Eke FU. Malaria-induced renal damage: facts and myths. *Pediatr Nephrol*. 2007;22:626–637.
92. Cohen SD, Kopp JB, Kimmel PL. Kidney diseases associated with human immunodeficiency virus infection. *N Engl J Med*. 2017;377:2363–2374.
93. Rao TK, Filippone EJ, Nicastri AD, et al. Associated focal and segmental glomerulosclerosis in the acquired immunodeficiency syndrome. *N Engl J Med*. 1984;310:669–673.
94. Pardo V, Aldana M, Colton RM, et al. Glomerular lesions in the acquired immunodeficiency syndrome. *Ann Intern Med*. 1984;101:429–434.
95. Gardenswartz MH, Lerner CW, Seligson GR, et al. Renal disease in patients with AIDS: a clinicopathologic study. *Clin Nephrol*. 1984;21:197–204.
96. Hory B, Bresson C, Lorge J, et al. Correspondence. *Am J Kidney Dis*. 1988;12:169.
97. Binda ki Muaka P, Pakasa-Muyulu N, Mbensa-Mansabi L. Abstracts of the 9th Congress of the International Pediatric Nephrology Association. *Pediatr Nephrol*. 1992;6:C154.
98. Bates WD, Muller N, van de Wal BW, et al. HIV-associated nephropathy-an initial presentation in an HIV-positive patient. *S Afr Med J*. 1994;84:223–224.
99. Fabian J, Naicker S. HIV and kidney disease in sub-Saharan Africa. *Nat Rev Nephrol*. 2009;5:591–598.
100. Rosenberg AZ, Naicker S, Winkler CA, et al. HIV-associated nephropathies: epidemiology, pathology, mechanisms and treatment. *Nat Rev Nephrol*. 2015;11:150–160.
101. Kopp JB, Nelson GW, Sampath K, et al. APOL1 genetic variants in focal segmental glomerulosclerosis and HIV-associated nephropathy. *J Am Soc Nephrol*. 2011;22:2129–2137.
102. Kasembeli AN, Duarte R, Ramsay M, et al. APOL1 risk variants are strongly associated with HIV-associated nephropathy in black South Africans. *J Am Soc Nephrol*. 2015.
103. Naicker S, Rahmanian S, Kopp JB. HIV and chronic kidney disease. *Clin Nephrol*. 2015;83:32–38.
104. Limou S, Nelson GW, Lecordier L, et al. Sequencing rare and common APOL1 coding variants to determine kidney disease risk. *Kidney Int*. 2015.
105. Koech M, Owiti M, Owino-Ong'or W, et al. Absence of HIV-associated nephropathy among antiretroviral naïve adults with persistent albuminuria in Western Kenya. *Kidney Int Rep*. 2017;2: 159–164.
106. Lucas GM, Eustace JA, Sozio S, et al. Highly active antiretroviral therapy and the incidence of HIV-1-associated nephropathy: a 12-year cohort study. *AIDS*. 2004;18:541–546.
107. Szczech LA, Gupta SK, Habash R, et al. The clinical epidemiology and course of the spectrum of renal diseases associated with HIV infection. *Kidney Int*. 2004;66:1145–1152.
108. Reid A, Stohr W, Walker AS, et al. Severe renal dysfunction and risk factors associated with renal impairment in HIV-infected adults in Africa initiating antiretroviral therapy. *Clin Infect Dis*. 2008;46:1271–1281.
109. Peters PJ, Moore DM, Mermin J, et al. Antiretroviral therapy improves renal function among HIV-infected Ugandans. *Kidney Int*. 2008;74:925–929.
110. Mpondo BC, Kalluvya SE, Peck RN, et al. Impact of antiretroviral therapy on renal function among HIV-infected Tanzanian adults: a retrospective cohort study. *PLoS ONE*. 2014;9:e89573.

111. Fabian J, Naicker S, Goetsch S, et al. The clinical and histological response of HIV-associated kidney disease to antiretroviral therapy in South Africans. *Nephrol Dial Transplant*. 2013;28:1543–1554.
112. *Global AIDS update 2016*. Geneva, Switzerland: UNAIDS; 2016.
113. Meintjes G, Moorhouse MA, Carmona S, et al. Adult antiretroviral therapy guidelines 2017. *South Afr J HIV Med*. 2017;18:1–24.
114. *AIDS by the numbers*. Geneva, Switzerland: Joint United Nations programme on HIV/AIDS; 2016.
115. Burnett RJ, Kramvis A, Dochez C, et al. An update after 16 years of hepatitis B vaccination in South Africa. *Vaccine*. 2012;30(suppl 3):C45–C51.
116. Combes B, Shorey J, Barrera A, et al. Glomerulonephritis with deposition of Australia antigen-antibody complexes in glomerular basement membrane. *Lancet*. 1971;2:234–237.
117. Bhimma R, Coovadia HM, Adhikari M. Hepatitis B virus-associated nephropathy in black South African children. *Pediatr Nephrol*. 1998;12:479–484.
118. Seggie J, Nathoo K, Davies PG. Association of hepatitis B (HBs) antigenaemia and membranous glomerulonephritis in Zimbabwean children. *Nephron*. 1984;38:115–119.
119. Wiggelinkhuizen J, Sinclair-Smith C. Membranous glomerulonephropathy in childhood. *S Afr Med J*. 1987;72:184–187.
120. Akinsola A, Olusanya O, Iyun AO, et al. Role of hepatitis Bs antigen in chronic glomerulonephritides in Nigerians. *Afr J Med Sci*. 1984;13:33–39.
121. Abdurrahman M, Aikhionbare H, Babaoye F, et al. Clinicopathological features of childhood nephrotic syndrome in northern Nigeria. *QJM*. 1990;75:563–576.
122. Bhimma R, Coovadia HM, Adhikari M, et al. The impact of the hepatitis B virus vaccine on the incidence of hepatitis B virus-associated membranous nephropathy. *Arch Pediatr Adolesc Med*. 2003;157:1025–1030.
123. Bates WD *Hepatitis-B-associated glomerular disease: a clinicopathological study of hepatitis b virus associated membranous glomerulonephritis in Namibian and South African children 1974-2005 and a comparison with hepatitis B associated membranous glomerulonephritis as well as idiopathic membranous glomerulonephritis in adults*. Cape Town: Stellenbosch University; 2011.
124. Moosa MR, Maree JD, Chirehwa MT, et al. Use of the 'Accountability for Reasonableness' approach to improve fairness in accessing dialysis in a middle-income country. *PLoS ONE*. 2016;11:e0164201.
125. Rodriguez-Iturbe B, Musser JM. The current state of poststreptococcal glomerulonephritis. *J Am Soc Nephrol*. 2008;19:1855–1864.
126. Nasr SH, Fidler ME, Valeri AM, et al. Postinfectious glomerulonephritis in the elderly. *J Am Soc Nephrol*. 2011;22:187–195.
127. Elzouki A, Amin F, Jaiswal O. Prevalence and pattern of renal disease in eastern Libya. *Arch Dis Child*. 1983;58:106–109.
128. Ben Maïz H, Abderrahim E, Ben Moussa F, et al. Epidemiology of glomerular diseases in Tunisia from 1975 to 2005. Influence of changes in healthcare and society. *Bull Acad Natl Med*. 2006;190:403–416.
129. Pinto SW, Sesso R, Vasconcelos E, et al. Follow-up of patients with epidemic poststreptococcal glomerulonephritis. *Am J Kidney Dis*. 2001;38:249–255.
130. Tiffin N, Hodkinson B, Okpechi I. Lupus in Africa: can we dispel the myths and face the challenges? *Lupus*. 2014;23:102–111.
131. Ameh O, Kengne A, Jayne D, et al. Standard of treatment and outcomes of adults with lupus nephritis in Africa: a systematic review. *Lupus*. 2016;25:1269–1277.
132. Appel GB, Contreras G, Dooley MA, et al. Mycophenolate mofetil versus cyclophosphamide for induction treatment of lupus nephritis. *J Am Soc Nephrol*. 2009;20:1103–1112.
133. Otieno L, McLigeyo S, Kayima J, et al. Management of lupus nephritis at the Kenyatta National Hospital. *East Afr Med J*. 1990;67:387–395.
134. Tannor E, Bates W, Moosa M. The clinical relevance of repeat renal biopsies in the management of lupus nephritis: a South African experience. *Lupus*. 2017.
135. Hotez PJ, Fenwick A. Schistosomiasis in Africa: an emerging tragedy in our new global health decade. *PLoS Negl Trop Dis*. 2009;3:e485.
136. Gryseels B. The relevance of schistosomiasis for public health. *Trop Med Parasitol*. 1989;40:134–142.
137. Barsoum RS. Schistosomiasis and the kidney. *Semin Nephrol*. 2003;23:34–41.
138. Barsoum RS, Bassily S, Soliman MM, et al. Renal amyloidosis and schistosomiasis. *Trans R Soc Trop Med Hyg*. 1979;73:367–374.
139. López M, Hoppe B. History, epidemiology and regional diversities of urolithiasis. *Pediatr Nephrol*. 2010;25:49–59.
140. Abboud O, Osman E, Musa A. The aetiology of chronic renal failure in adult Sudanese patients. *Ann Trop Med Parasitol*. 1989;83:411–414.
141. Elsharif ME, Elsharif EG. Causes of end-stage renal disease in Sudan: a single-center experience. *Saudi J Kidney Dis Transpl*. 2011;22:373–376.
142. El-Minshawy O. End-stage renal disease in the El-Minia governorate, upper Egypt: an epidemiological study. *Saudi J Kidney Dis Transpl*. 2010;22:1048–1054.
143. Arogundade FA, Hassan MO, Omotoso BA, et al. Spectrum of kidney diseases in Africa: malaria, schistosomiasis, sickle cell disease, and toxins. *Clin Nephrol*. 2016;86:S53–S60.
144. Naik RP, Derebail VK, Grams ME, et al. Association of sickle cell trait with chronic kidney disease and albuminuria in African Americans. *JAMA*. 2014;312:2115–2125.
145. Arogundade F, Sanusi A, Hassan M, et al. An appraisal of kidney dysfunction and its risk factors in patients with sickle cell disease. *Nephron Clin Pract*. 2011;118:c225–c231.
146. Nath KA, Hebbel RP. Sickle cell disease: renal manifestations and mechanisms. *Nat Rev Nephrol*. 2015;11:161–171.
147. Milne F, James S, Veriava Y. Malignant hypertension and its renal complications in black South Africans. *S Afr Med J*. 1989;78:164–167.
148. Meyers A, Whalley N, Martins M, et al. Hypertensive mechanisms in the indigenous (black) population of South Africa. *Nephrology*. 1998;4:S40–S47.
149. Katz IJ, Sofianou L, Butler O, et al. Recovery of renal function in Black South African patients with malignant hypertension: superiority of continuous ambulatory peritoneal dialysis over hemodialysis. *Perit Dial Int*. 2001;21:581–586.
150. Gold CH, Isaacson C, Levin J. The pathological basis of end-stage renal disease in blacks. *S Afr Med J*. 1982;61:263–265.
151. Freedman BI, Murea M. Target organ damage in African American hypertension: role of APOL1. *Curr Hypertens Rep*. 2012;14:21–28.
152. Freedman BI, Langefeld CD. The new era of APOL1-associated glomerulosclerosis. *Nephrol Dial Transplant*. 2012;27:1288–1291.
153. Fogo A, Breyer JA, Smith MC, et al. Accuracy of the diagnosis of hypertensive nephrosclerosis in African Americans: a report from the African American Study of Kidney Disease (AASK) Trial. AASK Pilot Study Investigators. *Kidney Int*. 1997;51:244–252.
154. Freedman BI, Sedor JR. Hypertension-associated kidney disease: perhaps no more. *J Am Soc Nephrol*. 2008;19:2047–2051.
155. Perneger TV, Whelton PK, Klag MJ, et al. Diagnosis of hypertensive end-stage renal disease: effect of patient's race. *Am J Epidemiol*. 1995;141:10–15.
156. Schlessinger SD, Tankersley MR, Curtis JJ. Clinical documentation of end-stage renal disease due to hypertension. *Am J Kidney Dis*. 1994;23:655–660.
157. Consortium HA, Rotimi C, Abayomi A, et al. Enabling the genomic revolution in Africa. *Science*. 2014;344:1346–1348.
158. Adedokun BO, Olopade CO, Olopade OI. Building local capacity for genomics research in Africa: recommendations from analysis of publications in Sub-Saharan Africa from 2004 to 2013. *Glob Health Action*. 2016;9:31026.
159. Osafo C, Raji YR, Burke D, et al. Human Heredity and Health (H3) in Africa Kidney Disease Research Network: a focus on methods in Sub-Saharan Africa. *Clin J Am Soc Nephrol*. 2015;10:2279–2287.
160. Osafo C, Raji YR, Olanrewaju T, et al. Genomic approaches to the burden of kidney disease in Sub-Saharan Africa: the Human Heredity and Health in Africa (H3Africa) Kidney Disease Research Network. *Kidney Int*. 2016;90:2–5.
161. Wonkam A, Makani J, Ofori-Aquah S, et al. Sickle cell disease and H3Africa: enhancing genomic research on cardiovascular diseases in African patients. *Cardiovasc J Afr*. 2015;26:50–55.
162. Treadwell MJ, Makani J, Ohene-Frempong K, et al. Stakeholder perspectives on public health genomics applications for sickle cell disease: a methodology for a Human Heredity and Health in Africa (H3Africa) qualitative research study. *OMICS*. 2017;21:323–332.
163. Jaouad IC, Elalaoui SC, Sbiti A, et al. Consanguineous marriages in Morocco and the consequence for the incidence of autosomal recessive disorders. *J Biosoc Sci*. 2009;41:575–581.
164. Bourquia A. Autosomal dominant polycystic kidney disease (ADPKD) in Morocco. Multicenter study about 308 families. *Nephrologie*. 2002;23:93–96, [Fr].
165. Laleye A, Awede B, Agboton B, et al. Autosomal dominant polycystic kidney disease in University Clinic of Nephrology and Haemodialysis of Cotonou: clinical and genetical findings. *Genet Couns*. 2012;23:435–445.

166. Fary Ka E, Seck SM, Niang A, et al. Patterns of autosomal dominant polycystic kidney diseases in black Africans. *Saudi J Kidney Dis Transpl.* 2010;21:81–86.
167. Seck SM, Guèye S, Diouf B. A new PKD-1 mutation discovered in a black African woman with autosomal polycystic kidney disease. *Nephrourol Mon.* 2013;5:769–772.
168. Yersin C, Bovet P, Wauters J, et al. Frequency and impact of autosomal dominant polycystic kidney disease in the Seychelles (Indian Ocean). *Nephrol Dial Transplant.* 1997;12:2069–2074.
169. Lombard EH, Kromberg JG, Thomson PD, et al. Autosomal recessive polycystic kidney disease. Evidence for high frequency of the gene in the Afrikaans-speaking population. *S Afr Med J.* 1989;76:321–323.
170. Lambie L, Amin R, Essop F, et al. Clinical and genetic characterization of a founder PKHD1 mutation in Afrikaners with ARPKD. *Pediatr Nephrol.* 2015;30:273–279.
171. Hildebrandt F, Zhou W. Nephronophthisis-associated ciliopathies. *J Am Soc Nephrol.* 2007;18:1855–1871.
172. Soliman NA, Hildebrandt F, Otto EA, et al. Clinical characterization and NPHP1 mutations in nephronophthisis and associated ciliopathies: a single center experience. *Saudi J Kidney Dis Transpl.* 2012;23:1090–1098.
173. Nagara M, Tiar A, Halim NB, et al. Mutation spectrum of primary hyperoxaluria type 1 in Tunisia: implication for diagnosis in North Africa. *Gene.* 2013;527:316–320.
174. Soliman NA, Nabhan MM, Abdelrahman SM, et al. Clinical spectrum of primary hyperoxaluria type 1: Experience of a tertiary center. *Nephrol Ther.* 2017;13:176–182.
175. Chaabouni HB, Ksantini M, M'Rad R, et al. MEFV mutations in Tunisian patients suffering from familial Mediterranean fever. *Semin Arthritis Rheum.* 2007;36:397–401.
176. Ben-Chetrit E. Familial Mediterranean fever (FMF) and renal AA amyloidosis-phenotype-genotype correlation, treatment and prognosis. *J Nephrol.* 2003;16:431–434.
177. Al-Haggar M, Bakr A, Tajima T, et al. Familial hypomagnesemia with hypercalciuria and nephrocalcinosis: unusual clinical associations and novel claudin16 mutation in an Egyptian family. *Clin Exp Nephrol.* 2009;13:288–294.
178. van der Merwe PD, Rensburg MA, Haylett WL, et al. Gitelman syndrome in a South African family presenting with hypokalaemia and unusual food cravings. *BMC Nephrol.* 2017;18:38.
179. Nkeh B, Samani NJ, Badenhorst D, et al. T594M variant of the epithelial sodium channel β-subunit gene and hypertension in individuals of African ancestry in South Africa. *Am J Hypertens.* 2003;16:847–852.
180. Jones ES, Owen EP, Rayner BL. The association of the R563Q genotype of the ENaC with phenotypic variation in Southern Africa. *Am J Hypertens.* 2012;25:1286–1291.
181. Dhanjal M, Owen E, Anthony J, et al. Association of pre-eclampsia with the R563Q mutation of the β-subunit of the epithelial sodium channel. *BJOG.* 2006;113:595–598.
182. Elhayek D, de Nanclares GP, Chouchane S, et al. Molecular diagnosis of distal renal tubular acidosis in Tunisian patients: proposed algorithm for Northern Africa populations for the ATP6V1B1, ATP6V0A4 and SCL4A1 genes. *BMC Med Genet.* 2013;14:119.
183. Lenkkeri U, Männikkö M, McCready P, et al. Structure of the gene for congenital nephrotic syndrome of the Finnish type (NPHS1) and characterization of mutations. *Am J Hum Genet.* 1999;64:51–61.
184. Gee HY, Saisawat P, Ashraf S, et al. ARHGDIA mutations cause nephrotic syndrome via defective RHO GTPase signaling. *J Clin Invest.* 2013;123:3243–3253.
185. Lunyera J, Mohottige D, Von Isenburg M, et al. CKD of uncertain etiology: a systematic review. *Clin J Am Soc Nephrol.* 2015;11:1–7.
186. Obrador GT, Schultheiss UT, Kretzler M, et al. Genetic and environmental risk factors for chronic kidney disease. *Kidney Int Suppl.* 2017;7:88–106.
187. Abid S, Hassen W, Achour A, et al. Ochratoxin A and human chronic nephropathy in Tunisia: is the situation endemic? *Hum Exp Toxicol.* 2003;22:77–84.
188. Kamel EG, El-Minshawy O. Environmental factors incriminated in the development of end stage renal disease in El-Minia Governorate, Upper Egypt. *Int J Nephrol Urol.* 2010;2:431–437.
189. Okpechi IG, Bello AK, Ameh OI, et al. Integration of care in management of CKD in resource-limited settings. *Semin Nephrol.* 2017;37:260–272.
190. De Broe ME, Benghanem Gharbi M, Zamd M, et al. Why overestimate or underestimate chronic kidney disease when correct estimation is possible? *Nephrol Dial Transplant.* 2017;32:ii136–ii141.
191. Ekiti ME, Zambo J-B, Assah FK, et al. Chronic kidney disease in sugarcane workers in Cameroon: a cross-sectional study. *BMC Nephrol.* 2018;19:10.
192. Diana NE, Naicker S. Update on current management of chronic kidney disease in patients with HIV infection. *Int J Nephrol Renovasc Dis.* 2016;9:223–234.
193. George C, Mogueo A, Okpechi I, et al. Chronic kidney disease in low-income to middle-income countries: the case for increased screening. *BMJ Glob Health.* 2017;2:e000256.
194. Bonita R, de Courten M, Dwyer T, et al *Surveillance of risk factors for noncommunicable diseases: The WHO STEPwise approach.* Geneva, Switzerland: World Health Organization; 2001.
195. Arogundade FA, Barsoum RS. CKD prevention in Sub-Saharan Africa: a call for governmental, nongovernmental, and community support. *Am J Kidney Dis.* 2008;51:515–523.
196. Anand S, Bitton A, Gaziano T. The gap between estimated incidence of end-stage renal disease and use of therapy. *PLoS ONE.* 2013;8:e72860.
197. Barsoum RS. Chronic kidney disease in the developing world. *N Engl J Med.* 2006;354:997–999.
198. Barsoum RS. Burden of chronic kidney disease: North Africa. *Kidney Int Suppl.* 2013;3:164–166.
199. El Nahas M. The global challenge of chronic kidney disease. *Kidney Int.* 2005;68:2918–2929.
200. Schieppati A, Remuzzi G. Chronic renal diseases as a public health problem: epidemiology, social, and economic implications. *Kidney Int Suppl.* 2005;S7–S10.
201. Liyanage T, Ninomiya T, Jha V, et al. Worldwide access to treatment for end-stage kidney disease: a systematic review. *Lancet.* 2015;385:1975–1982.
202. Arogundade F, Sanusi A, Hassan M, et al. The pattern, clinical characteristics and outcome of ESRD in Ile-Ife, Nigeria: Is there a change in trend? *Afr Health Sci.* 2011;11:594–601.
203. Davids MR, Caskey FJ, Young T, et al. Strengthening renal registries and ESRD research in Africa. *Semin Nephrol.* 2017;37:211–223.
204. Naicker S. End-stage renal disease in Sub-Saharan Africa. *Kidney Int Suppl.* 2013;3:161–163.
205. Benghanem Gharbi M. Renal replacement therapies for end-stage renal disease in North Africa. *Clin Nephrol.* 2010;74:S17–S19.
206. Davids MR, Balbir Singh GK, Marais N, et al. *South African Renal Registry Annual Report 2014.* Cape Town: South African Renal Society; 2016.
207. Davids MR, Marais N, Balbir Singh GK, et al *South African Renal Registry Annual Report 2013.* Cape Town: South African Renal Society; 2015.
208. Davids MR, Marais N, Jacobs JC *South African Renal Registry Annual Report 2012.* Cape Town: South African Renal Society; 2014.
209. Davids MR, Eastwood JB, Selwood NH, et al. A renal registry for Africa: first steps. *Clin Kidney J.* 2016;9:162–167.
210. *United States Renal Data System 2017 Annual Data Report.* Bethesda, MD: National Institutes of Health, National Institute of Diabetes and Digestive and Kidney Diseases; 2017.
211. Mushi L, Krohn M, Flessa S. Cost of dialysis in Tanzania: evidence from the provider's perspective. *Health Econ Rev.* 2015;5:28.
212. Halle MP, Jimkap NN, Kaze FF, et al. Cost of care for patients on maintenance haemodialysis in public facilities in Cameroon. *Afr J Nephrol.* 2017;20:230–237.
213. Ashuntantang G, Osafo C, Olowu WA, et al. Outcomes in adults and children with end-stage kidney disease requiring dialysis in sub-Saharan Africa: a systematic review. *Lancet Glob Health.* 2017;5:e408–e417.
214. Okaka EI, Naidoo S, Ahmed MM, et al. Quality of life in patients on continuous ambulatory peritoneal dialysis in an African setting. *Saudi J Kidney Dis Transpl.* 2015;26:631–637.
215. Okaka EI, Davies M, Ahmed M, et al. Impact of socio-economic factors on quality of life in patients on continuous ambulatory peritoneal dialysis in an African setting. *West Afr J Med.* 2014;33:125–129.
216. Tannor EK, Archer E, Kapembwa K, et al. Quality of life in patients on chronic dialysis in South Africa: a comparative mixed methods study. *BMC Nephrol.* 2017;18:4.
217. Masina T, Chimera B, Kamponda M, et al. Health related quality of life in patients with end stage kidney disease treated with haemodialysis in Malawi: a cross sectional study. *BMC Nephrol.* 2016;17:61.

218. Kamau E, Kayima J, Otieno C, et al. Health related quality of life of patients on maintenance haemodialysis at Kenyatta National Hospital. *East Afr Med J.* 2012;89:75–81.
219. El Filali A, Bentata Y, Ada N, et al. Depression and anxiety disorders in chronic hemodialysis patients and their quality of life: a cross-sectional study about 106 cases in the northeast of morocco. *Saudi J Kidney Dis Transpl.* 2017;28:341–348.
220. Bouidida B, Rhou H, Ezaitouni F, et al. Translation, cultural adaptation and validation of the Kidney Disease Quality of Life–Short Form 1.3 in an African country. *Transplant Proc.* 2014;46:1295–1301.
221. Abdelfatah M, Shawky S, Azzab E. Quality of life in dialysis versus kidney transplantation. *Afr J Nephrol.* 2010;14:52–60.

BOARD REVIEW QUESTIONS

1. A systematic review on the population prevalence of CKD in sub-Saharan Africa estimated the prevalence at 13.9%. The large Maremar study in Morocco included over ten thousand adults and reported a population prevalence of 5.1%. What is the likeliest reason for this lower prevalence?
   a. CKD is less common in North Africa.
   b. HIV infection drives CKD in sub-Saharan Africa.
   c. Maremar did not include stages 1 and 2 CKD.
   d. Maremar required that the criterion of chronicity be satisfied.
   e. Different equations were used to estimate GFR.

   Answer: d

   Rationale: Many prevalence studies are cross-sectional and suffer from the methodological weakness that abnormal findings are not repeated to satisfy the criterion of chronicity for the diagnosis of CKD. Among African studies, the Maremar study is a notable exception. Another recent study in sugarcane workers in Cameroon also included the element of chronicity and reported a prevalence of 3.4%.

2. Black patients are often labeled as having "hypertensive renal disease" when they present with advanced CKD, small kidneys and hypertension. There is often no clear history of hypertension preceding the renal disease. In the absence of another clear cause, what is the likely etiology of their renal disease?
   a. Essential hypertension with hypertensive nephrosclerosis
   b. Glomerular disease related to APOL1 gene variants
   c. Hypertension due to gene variants in the renin-angiotensin-aldosterone system
   d. CKDu related to repeated heat exposure and dehydration
   e. CKDu related to environmental toxins

   Answer: b

   Rationale: It is now thought that many of these patients have glomerular disease related to the presence of "African" renal-risk variants of the APOL1 gene. A primary renal diagnosis of hypertensive renal disease should be considered only when there is hypertension known to precede renal dysfunction, left ventricular hypertrophy, proteinuria <2 g/day, and no evidence of other renal diseases. If there is any doubt, patients should rather be labeled as "CKD/ESRD–cause unknown".

3. Which African country has the highest utilization of peritoneal dialysis as a treatment modality for end-stage renal disease?
   a. Morocco
   b. Egypt
   c. Sudan
   d. Kenya
   e. South Africa

   Answer: e

   Rationale: South Africa had 10,360 patients on renal replacement therapy in December 2015, a prevalence of 189 PMP. Hemodialysis was the treatment modality in 72.7%, peritoneal dialysis in 13.9%, and transplantation in 13.4%. In the public healthcare sector, which caters for 84% of the population, peritoneal dialysis was the modality for 29% of the patients.